



CS 225

Data Structures

Volodymyr Kindratenko

Slides courtesy of Wade Fagen-Ulmschneider

Every Data Structure So Far

	Unsorted Array	Sorted Array	Unsorted List	Sorted List	Binary Tree	BST	AVL
Find							
Insert							
Remove							
Traverse							

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Range-based Searches

Q: Consider points in 1D: $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$.
...what points fall in $[11, 42]$?

Tree construction:

Range-based Searches

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Tree construction:

— Balanced BST

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Q: Consider points in 1D: $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$.
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Tree construction:

- Balanced BST
- Leaf nodes are values
- Inner nodes are bounds

Range-based Searches

Balanced BSTs are useful structures for range-based and nearest-neighbor searches.

Q: Consider points in 1D: $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$.
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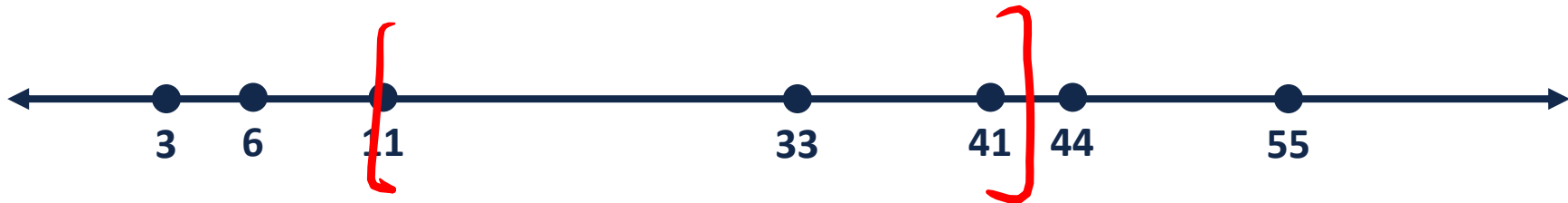
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*create sorted array
 $O(n \lg n)$*

Q: Consider points in 1D: $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$.
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Ex:



(11, 33, 41)

Range-based Searches

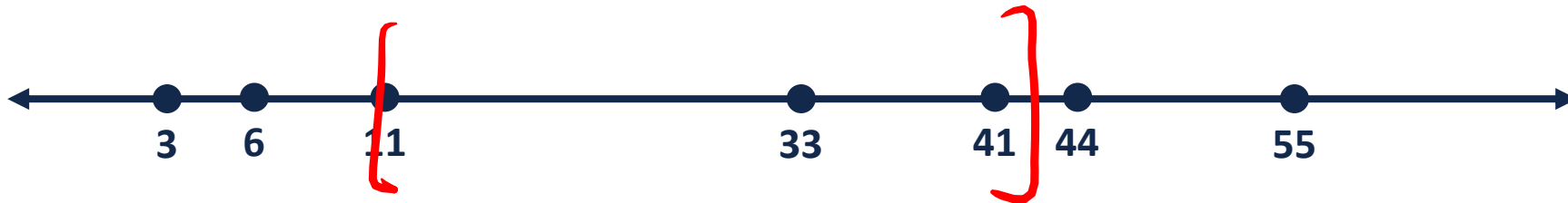
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find $O(\lg n)$

Ex:



$(11, 33, 41)$

Range-based Searches

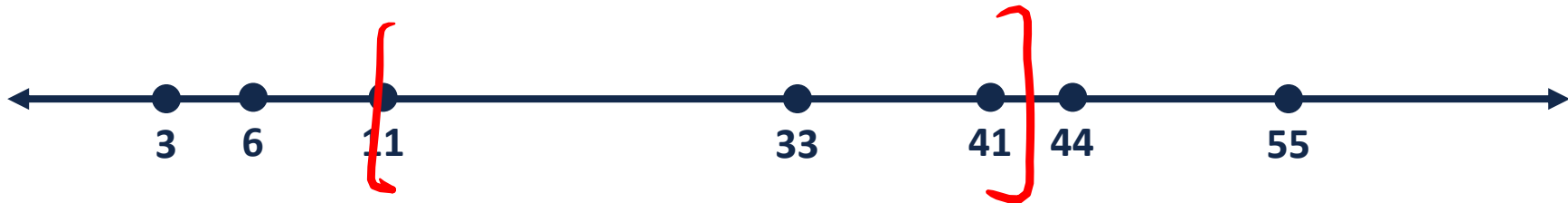
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find $O(\lg n)$
access k elements
 $O(k)$

Ex:



$(11, 33, 41)$

Range-based Searches

Balanced BSTs are useful structures for range-based and nearest-neighbor searches.

create sorted array
 $O(n \lg n)$

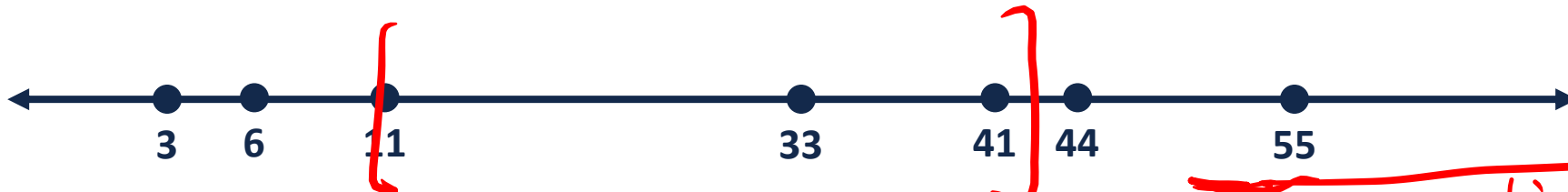
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running time (array)
 $O(n \lg n + k)$

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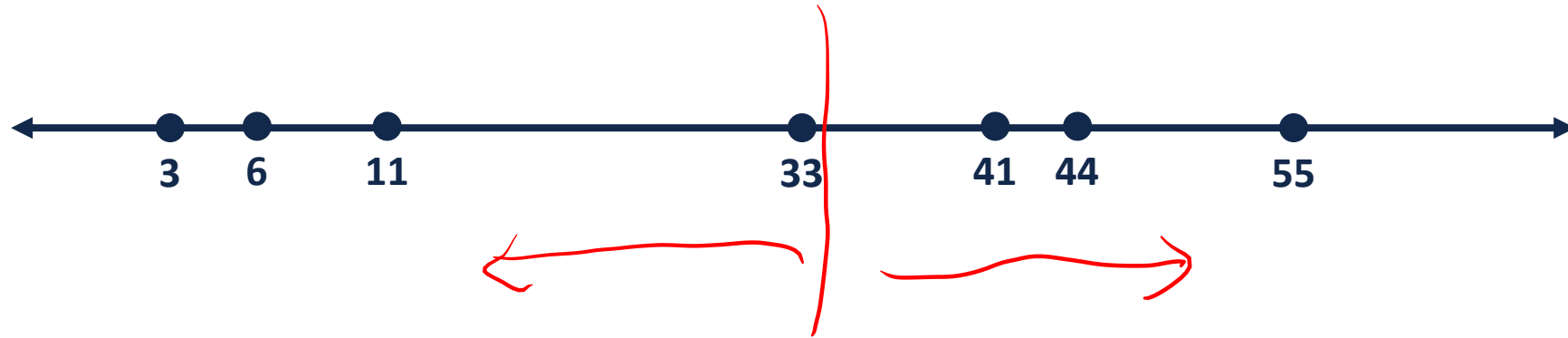
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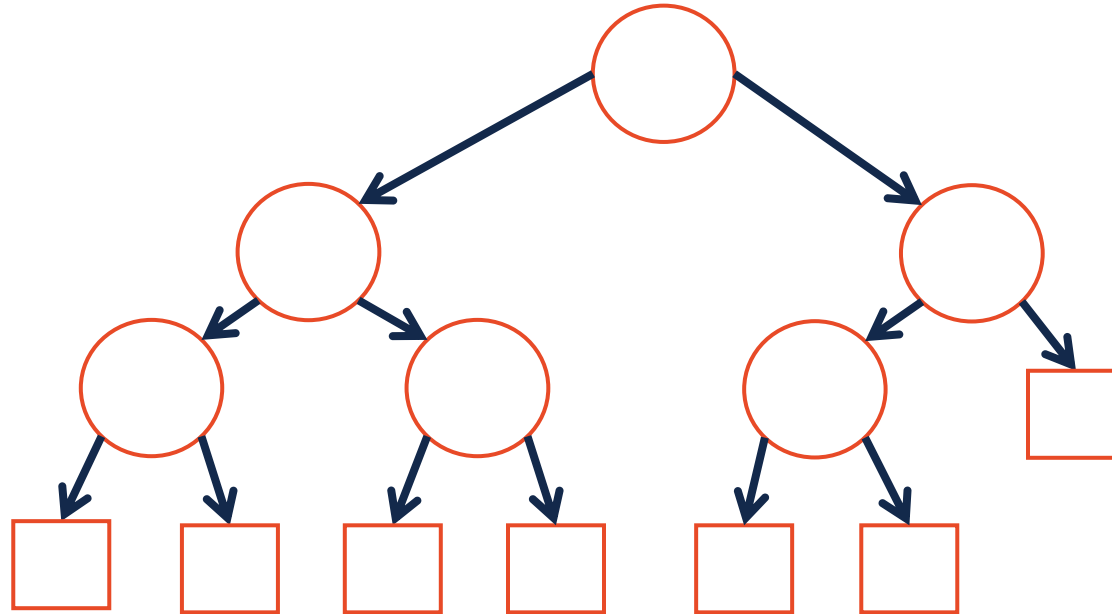
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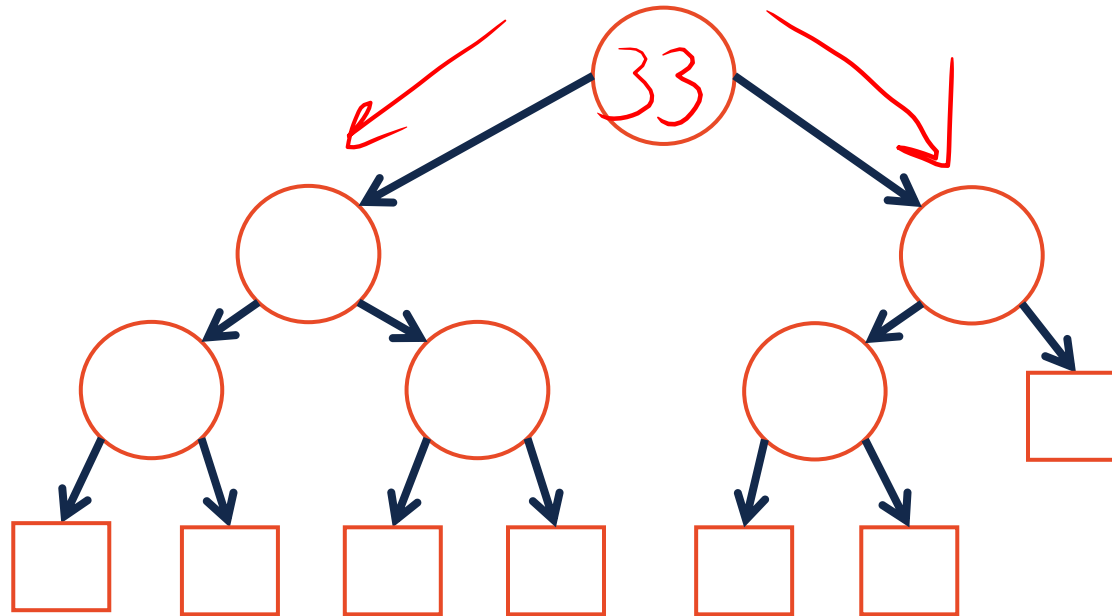
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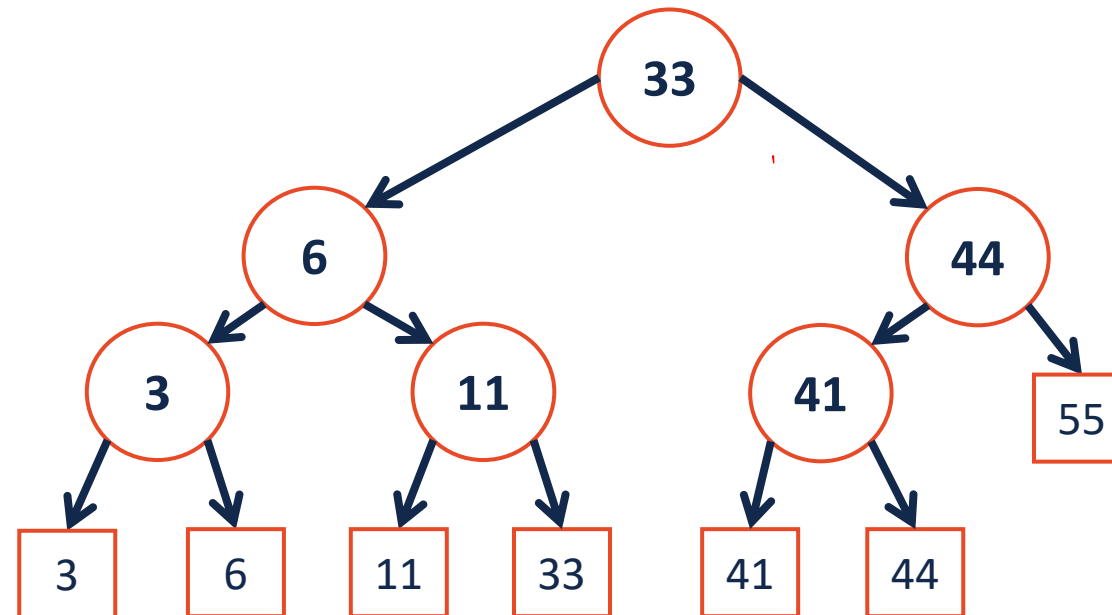
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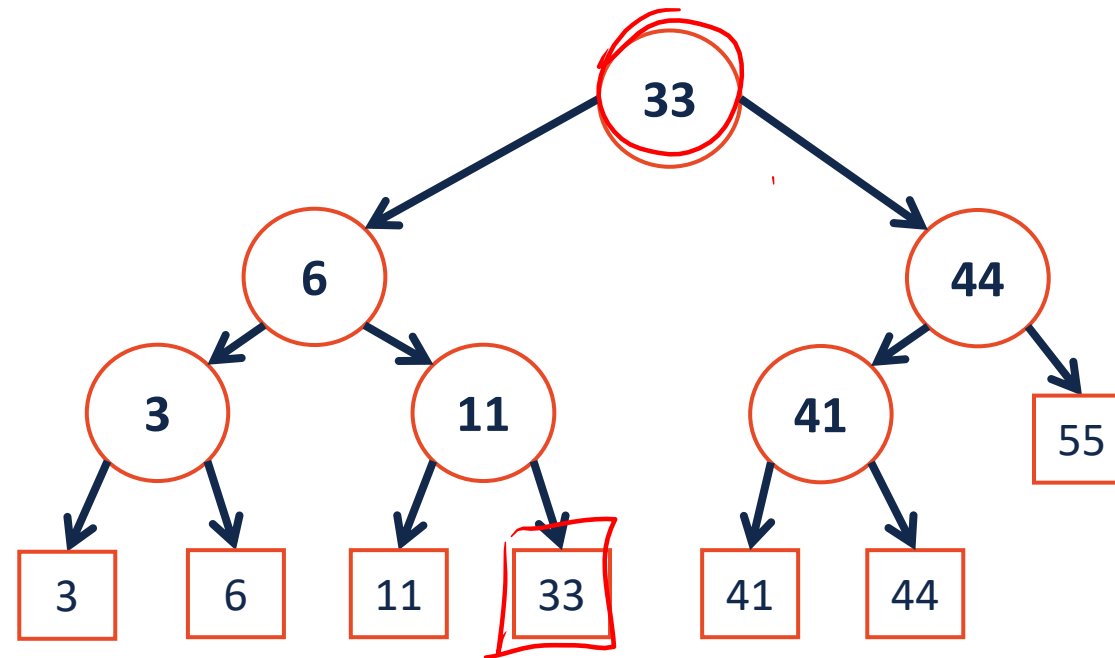
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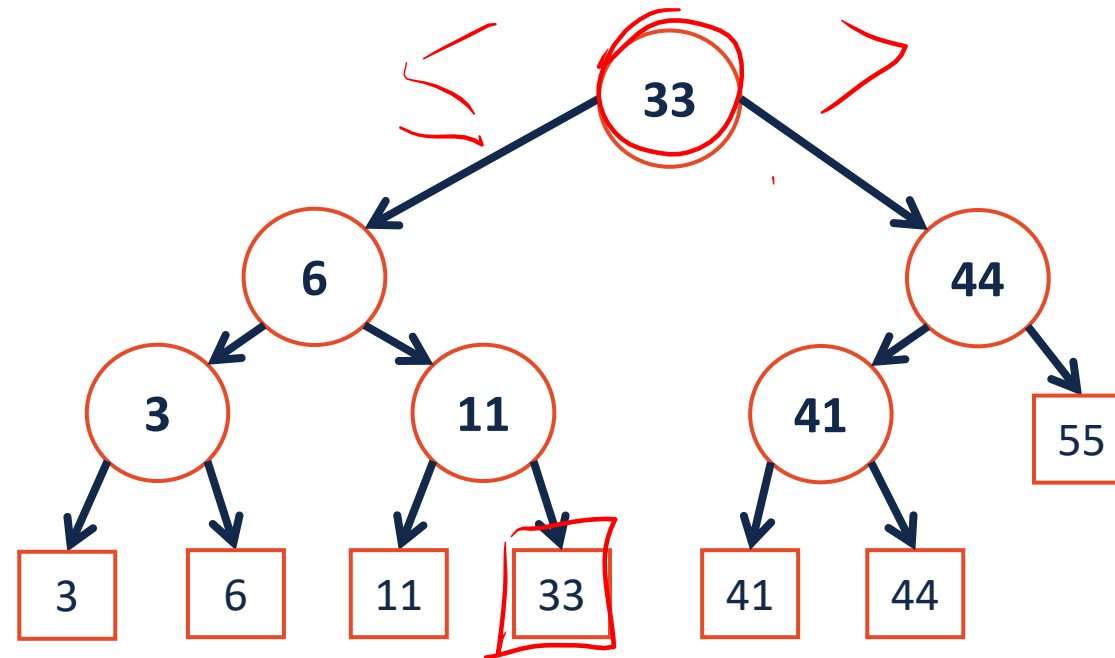
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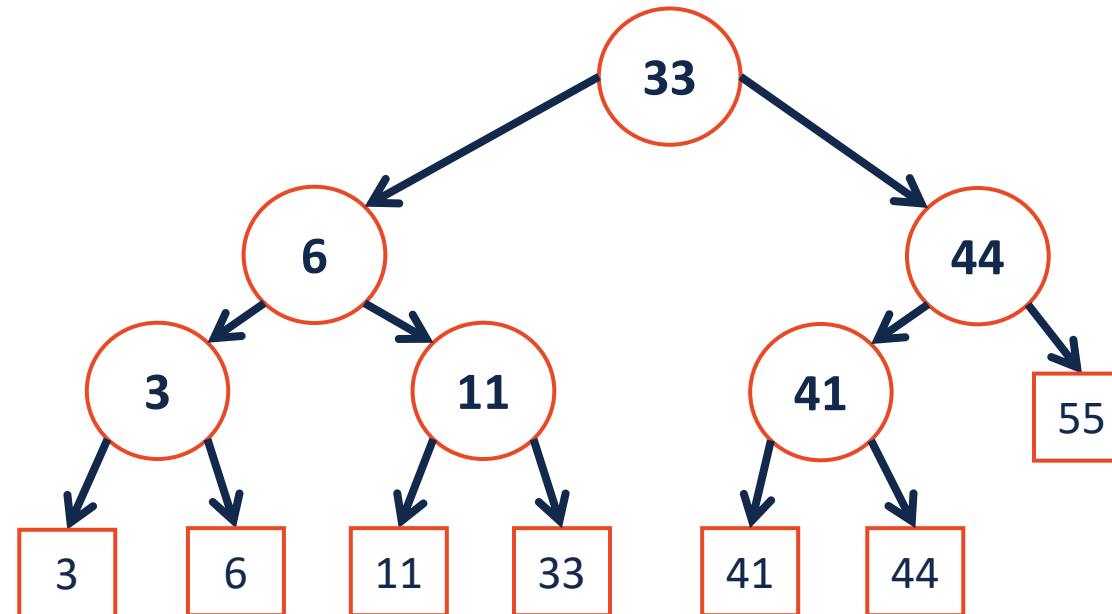


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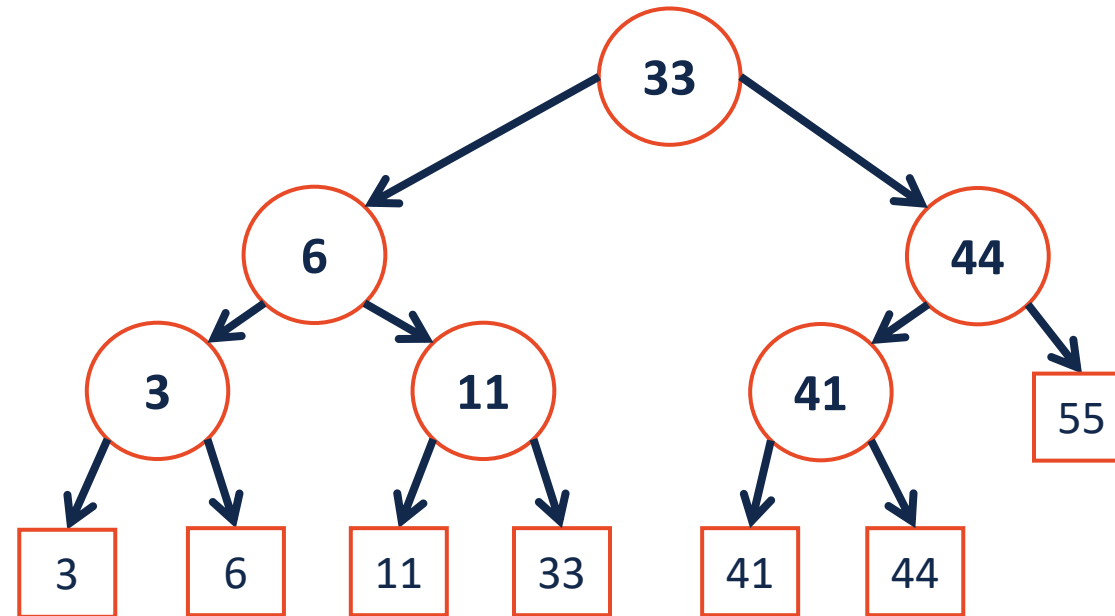
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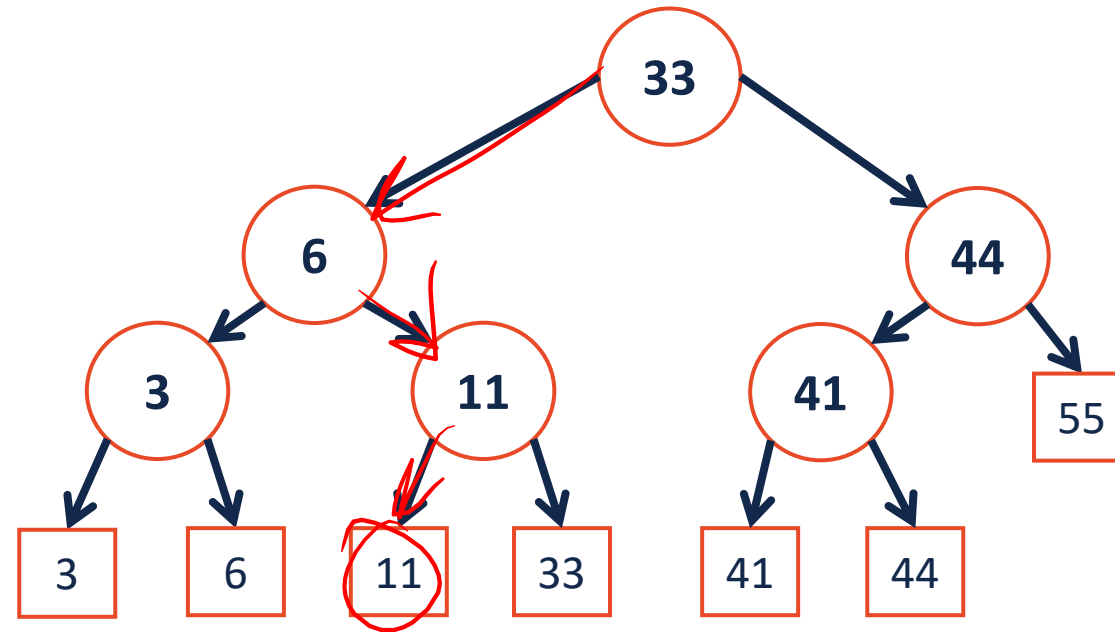
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$P_{low} = 11$
 $P_{high} = 42$

Range-based Searches

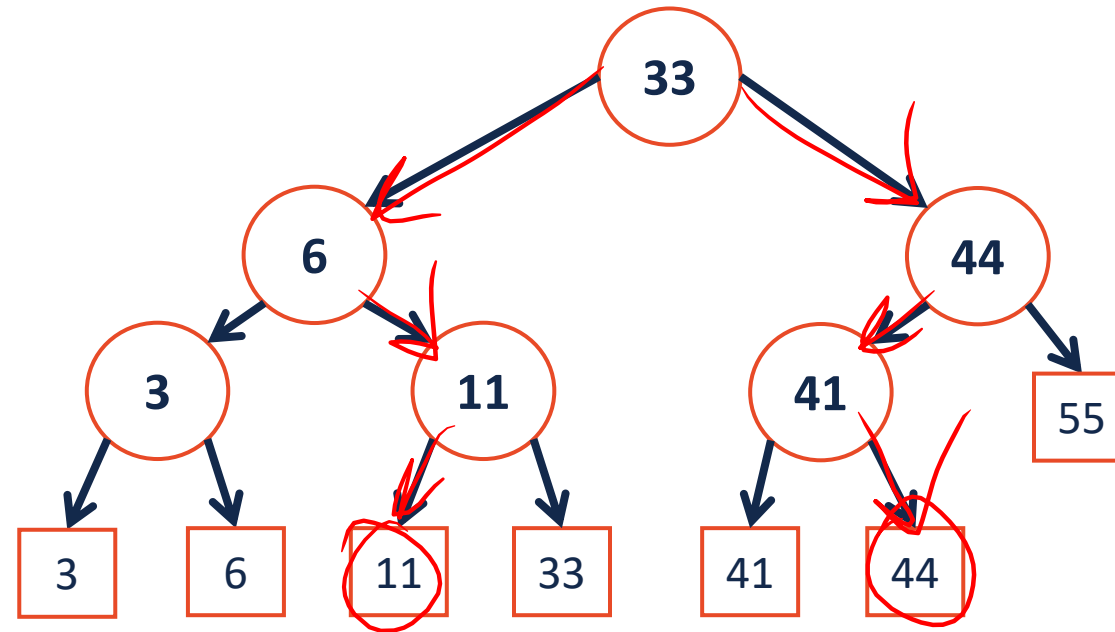
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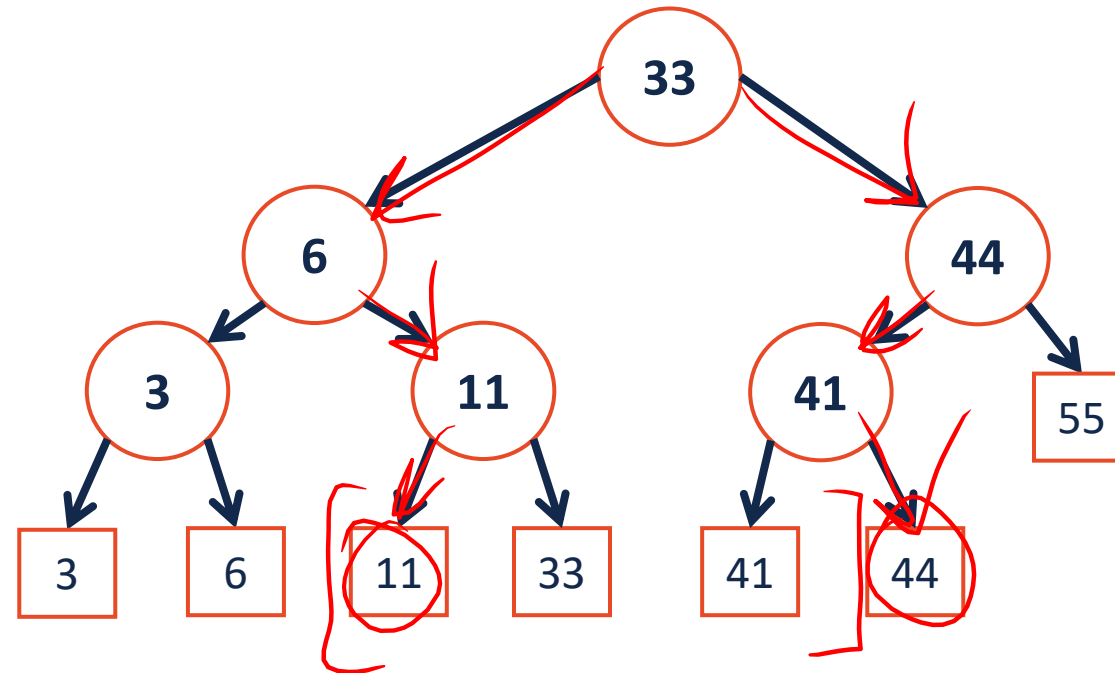
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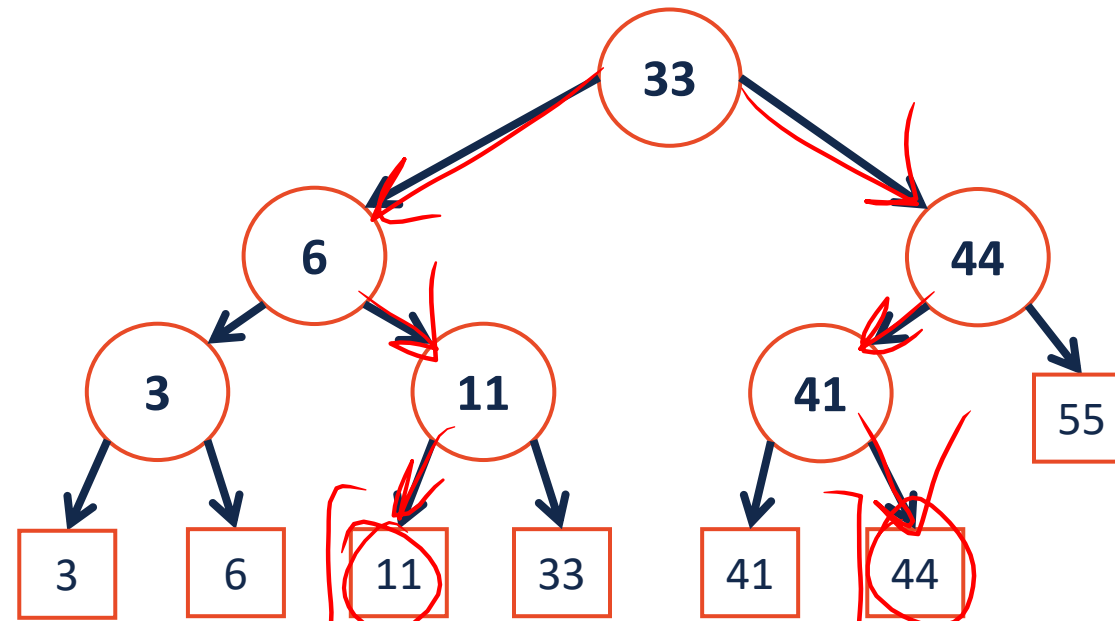
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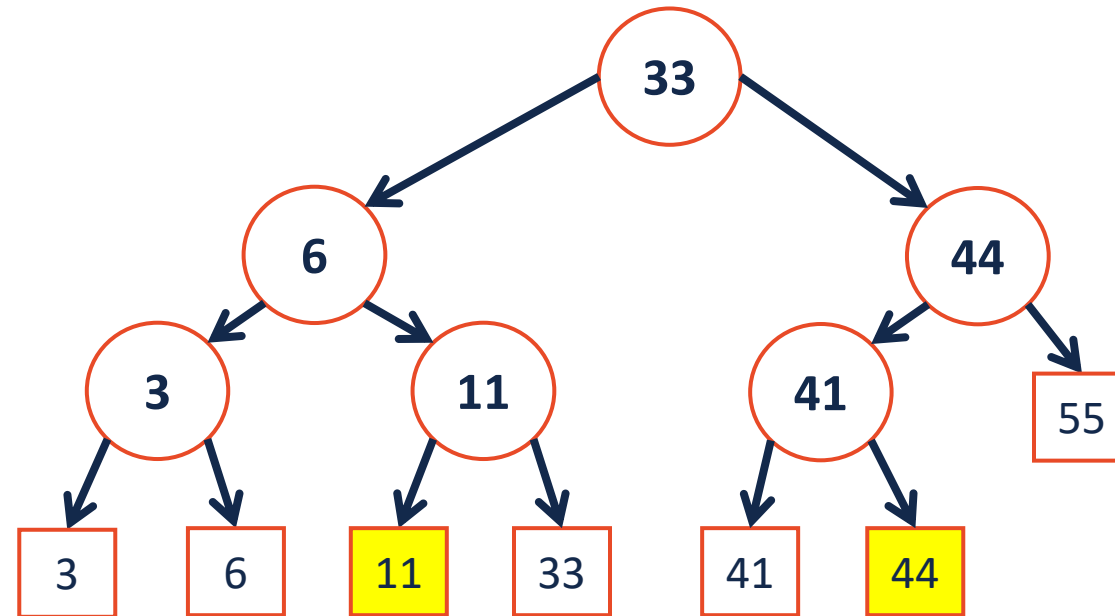
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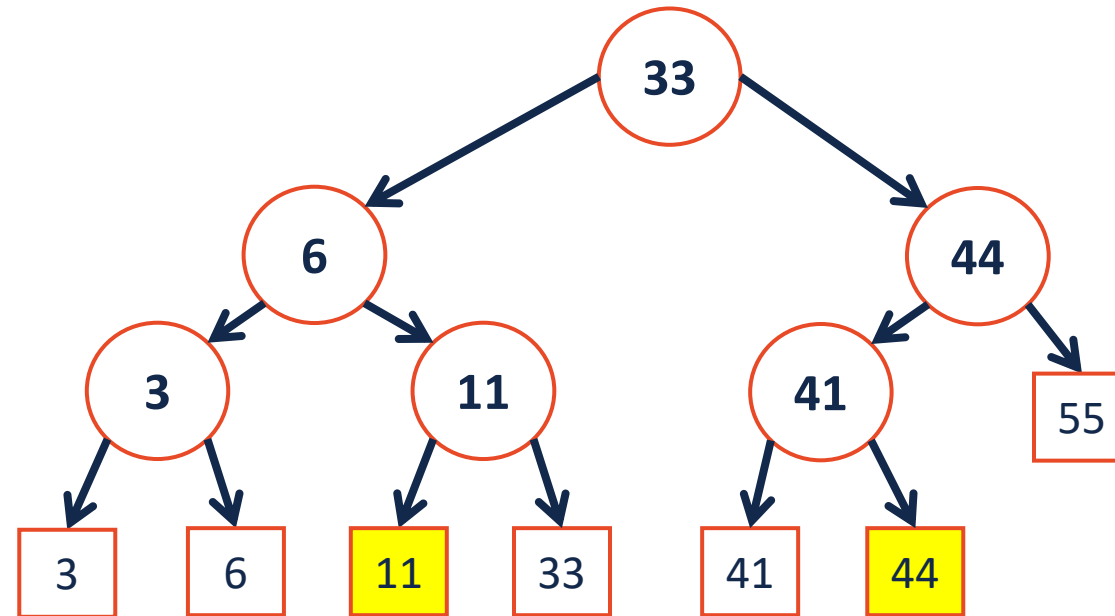
in-order traversal

Running Time



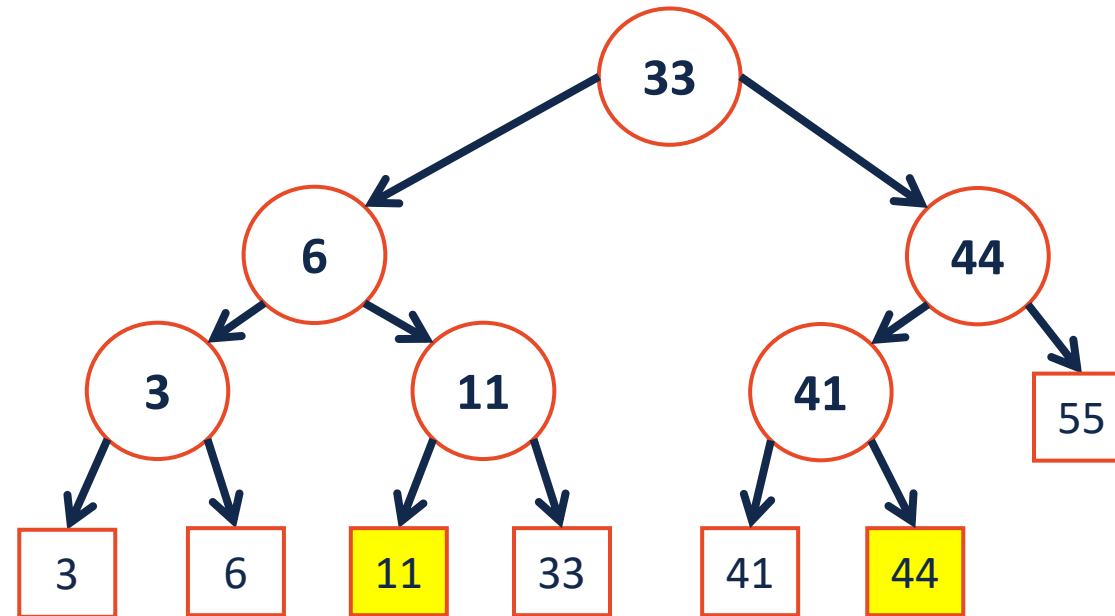
Running Time

insertion $O(n \lg n)$



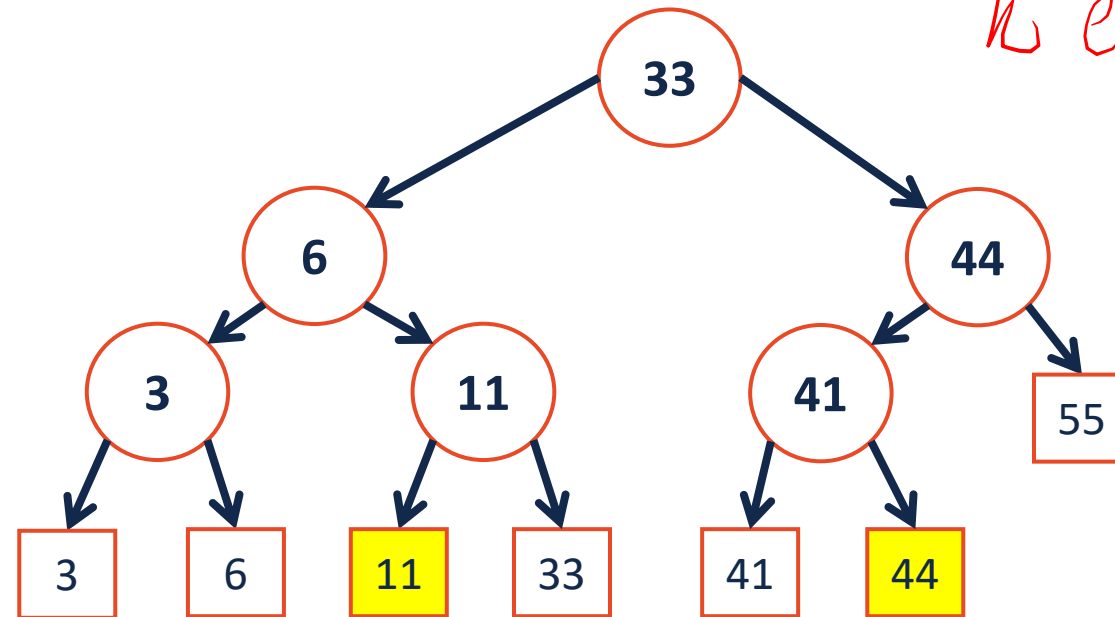
Running Time

insertion $O(n \lg n)$
find $2 * O(\lg n)$



Running Time

insertion $O(n \lg n)$
find $2 * O(\lg n)$
 k elements $O(k)$



create link list

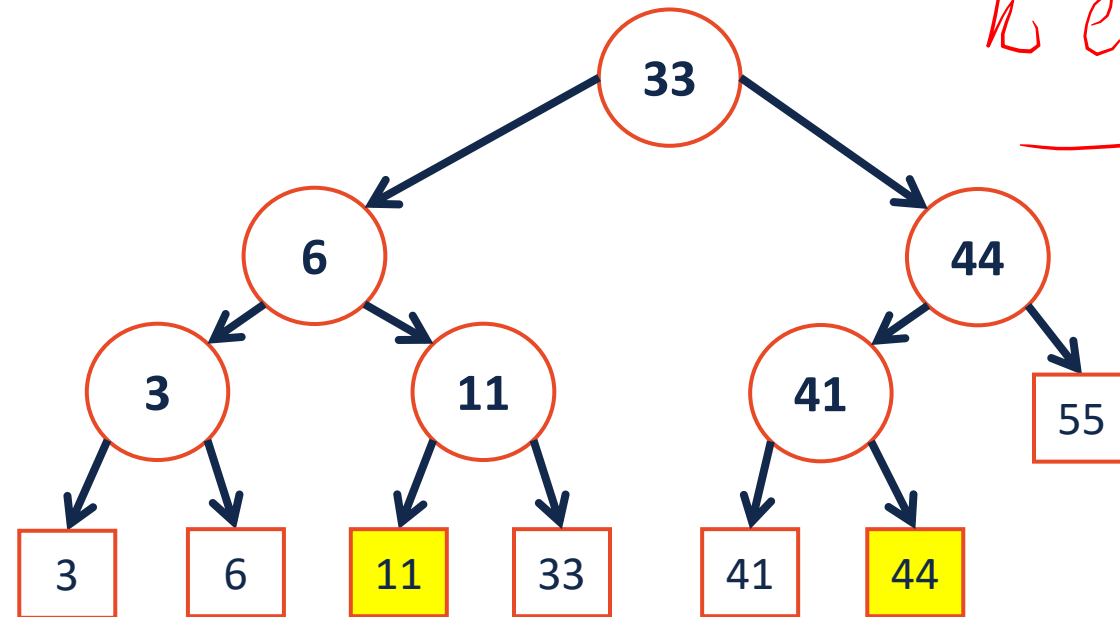
Running Time

insertion $O(n \lg n)$

find $2 * O(\lg n)$

k elements $O(k)$

$O(n \lg n + k)$



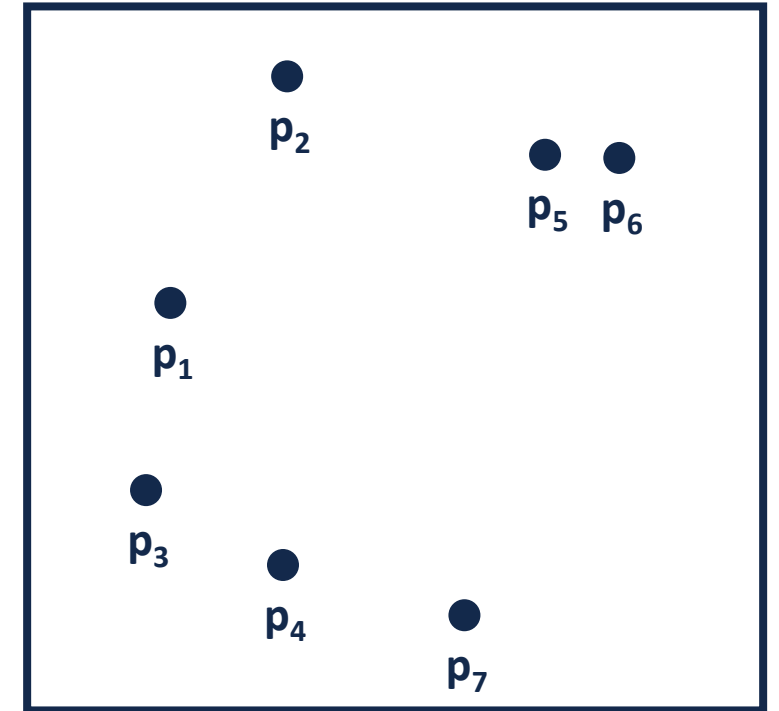
create link list

Range-based Searches

Consider points in 2D: $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$.

Q: What points are in the rectangle:
[$(x_1, y_1), (x_2, y_2)$]?

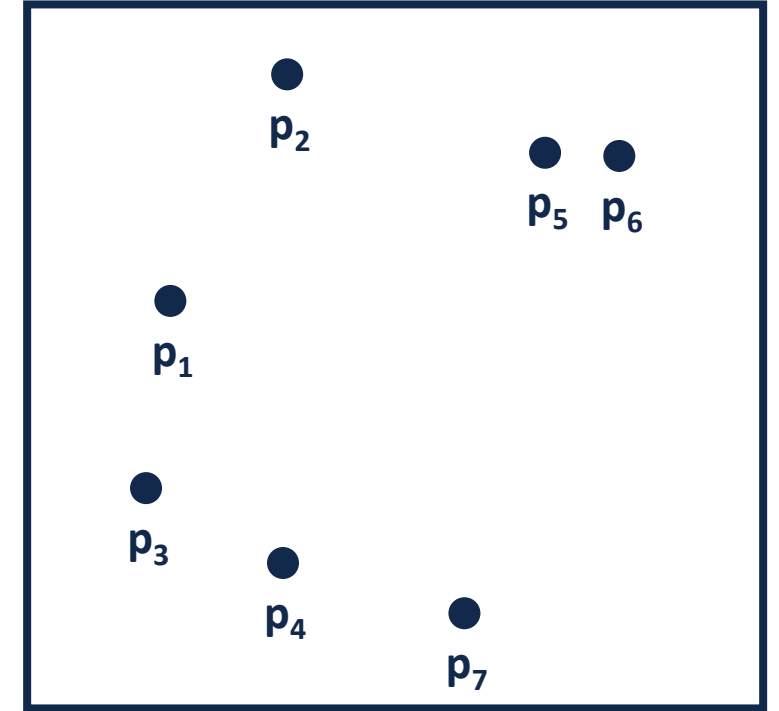
Q: What is the nearest point to (x_1, y_1) ?



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Consider points in 2D: $\mathbf{p} = \{p_1, p_2, \dots, p_n\}$.

Tree construction:

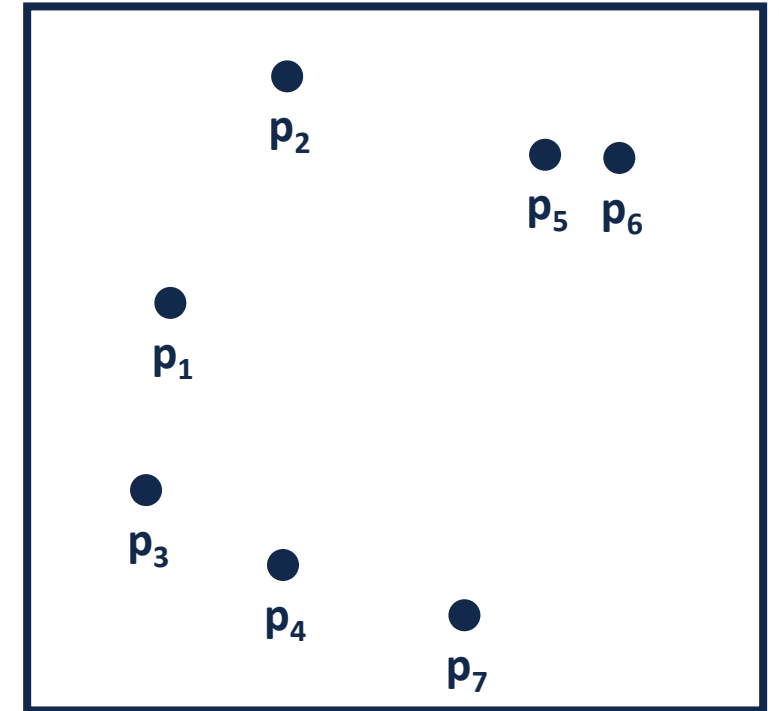


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Tree construction:

Balanced BST



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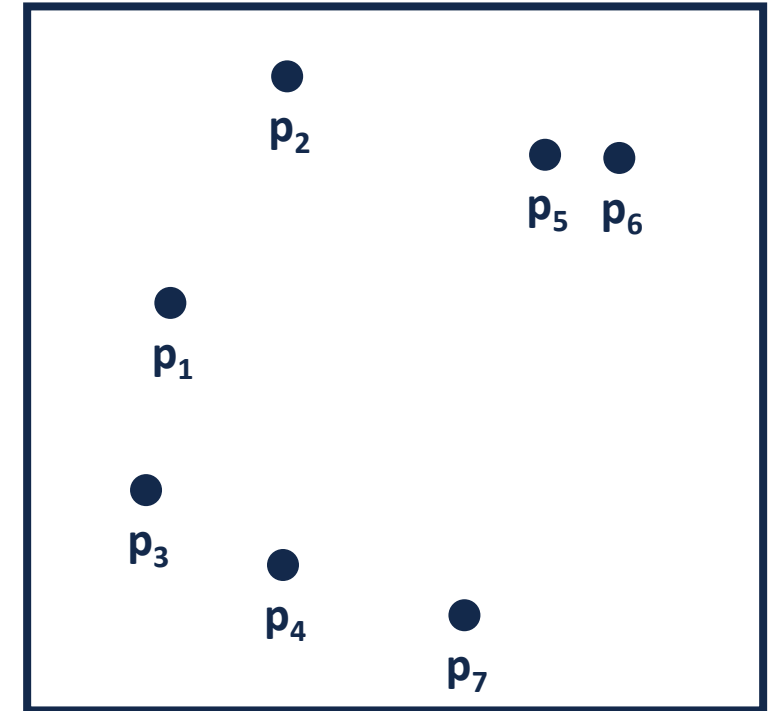
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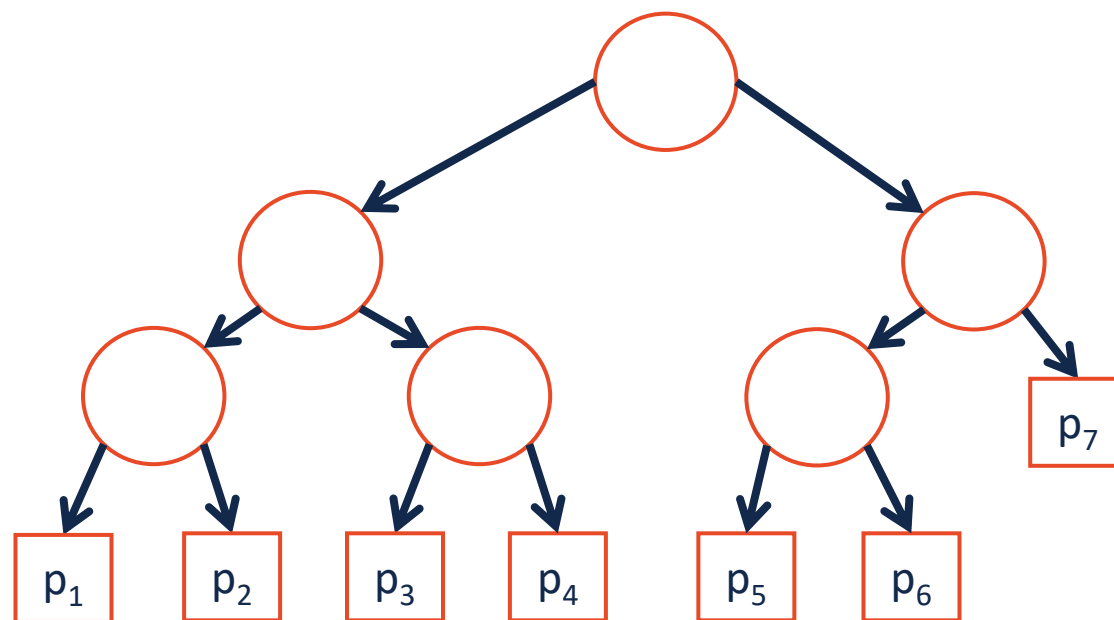
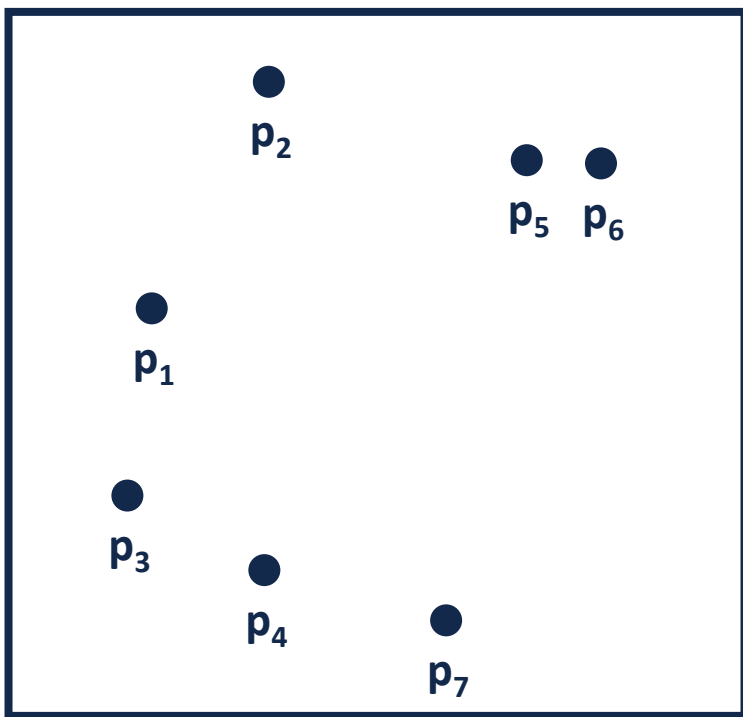
Balanced BST

— Inner nodes are partitions

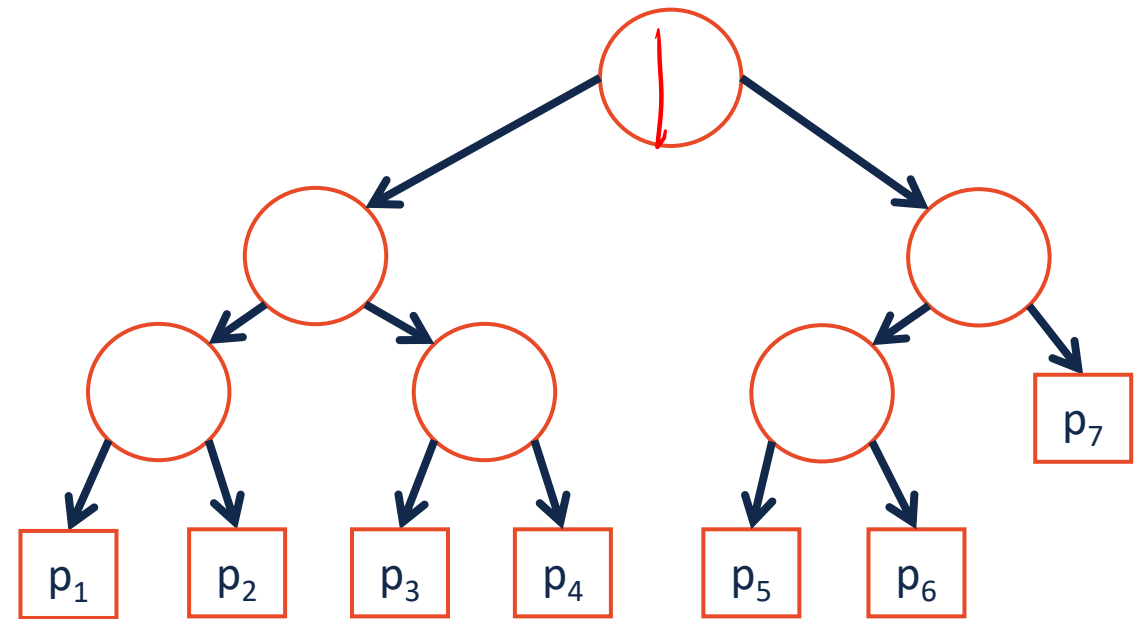
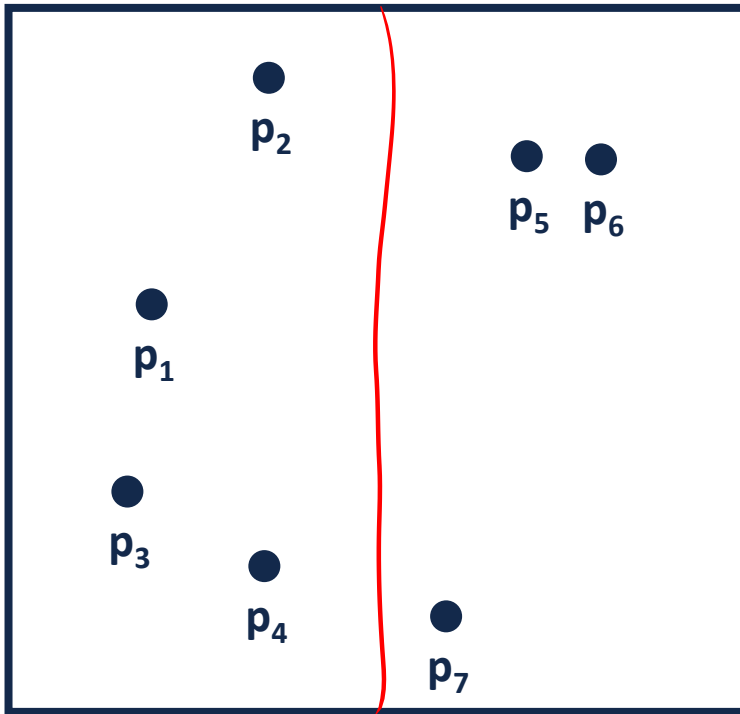
— Leaf nodes are points



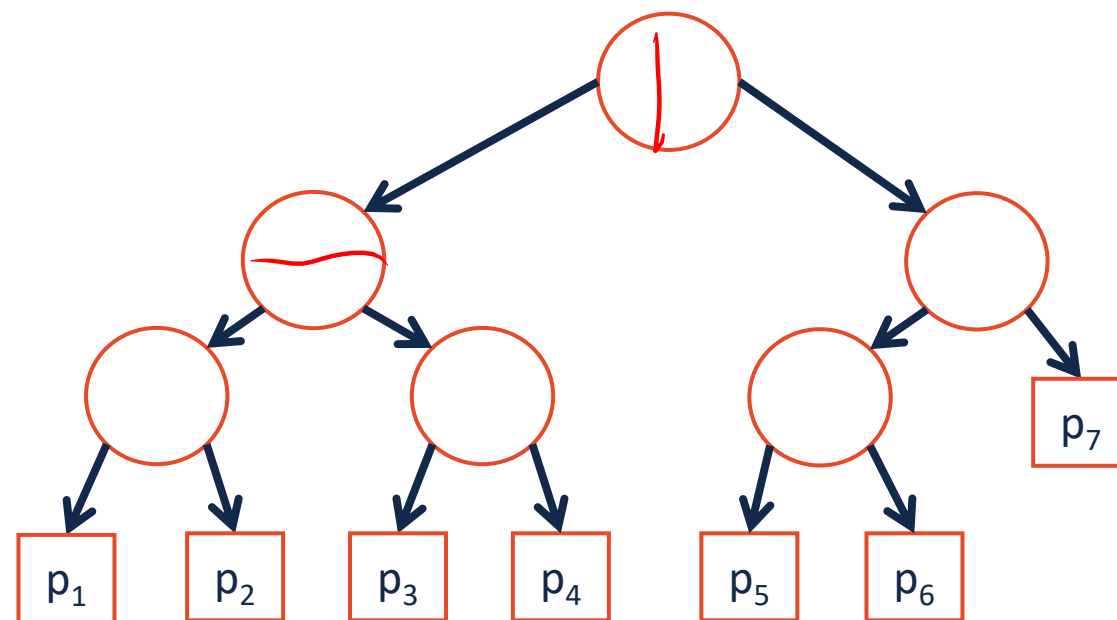
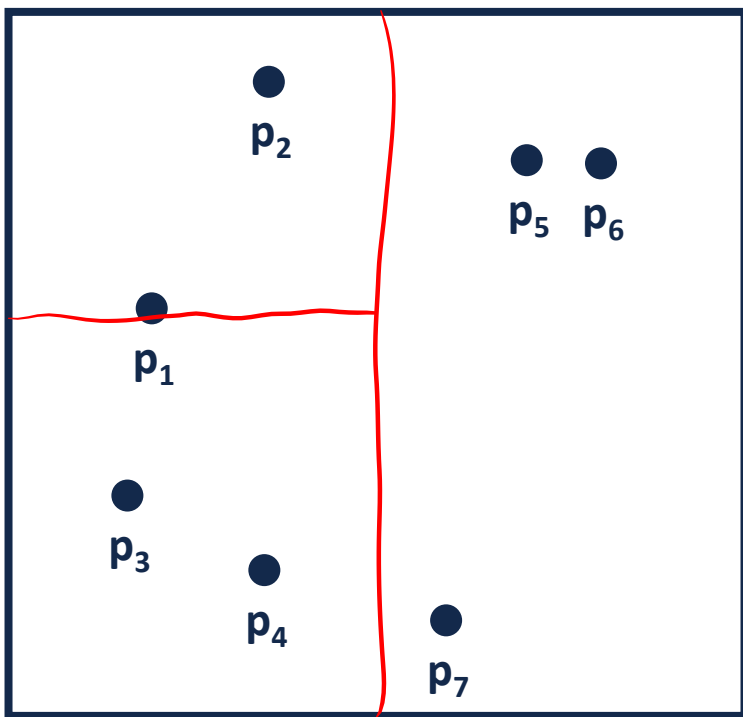
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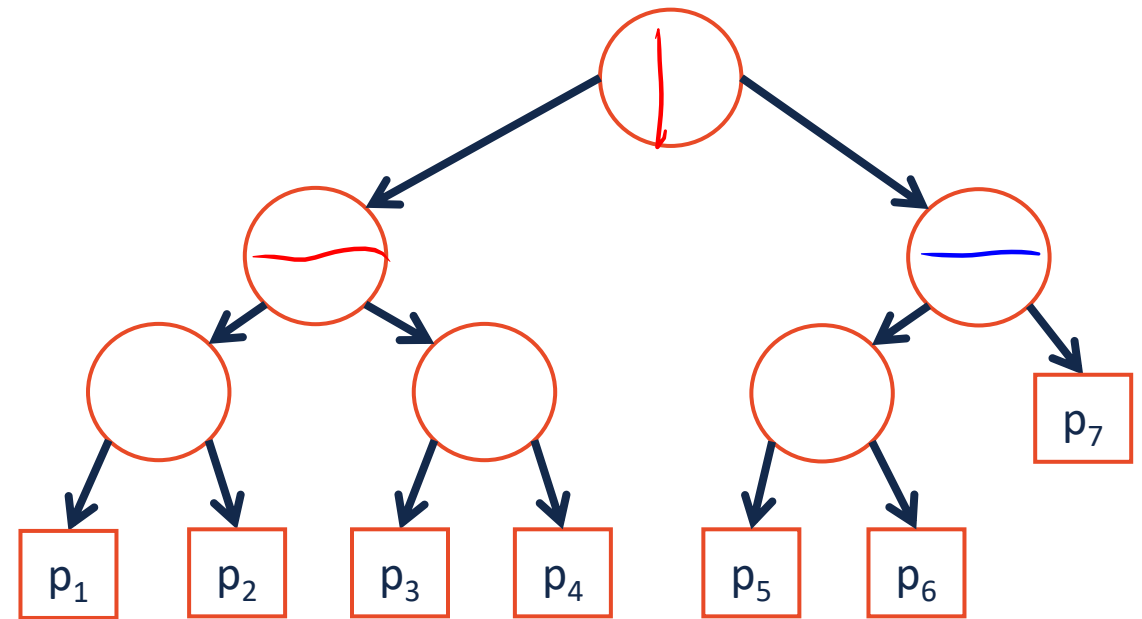
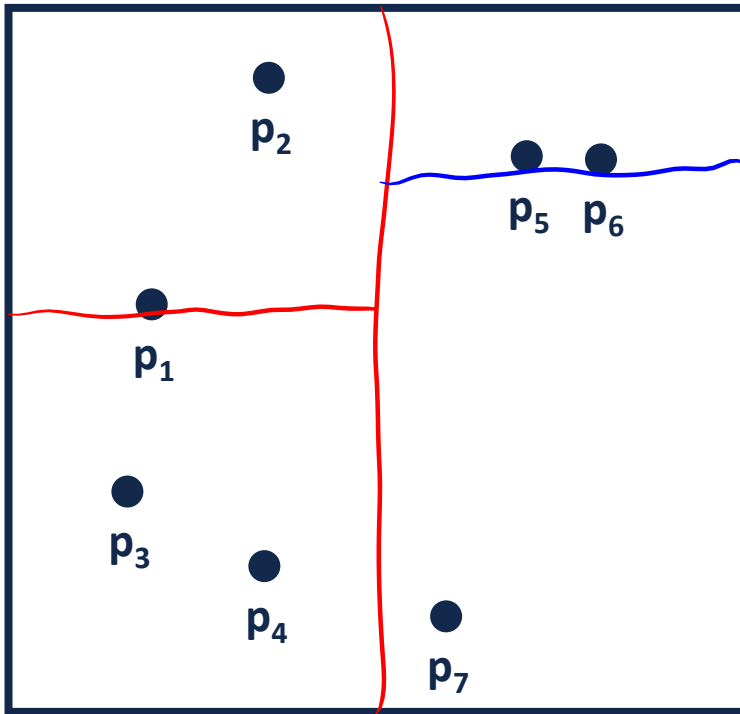
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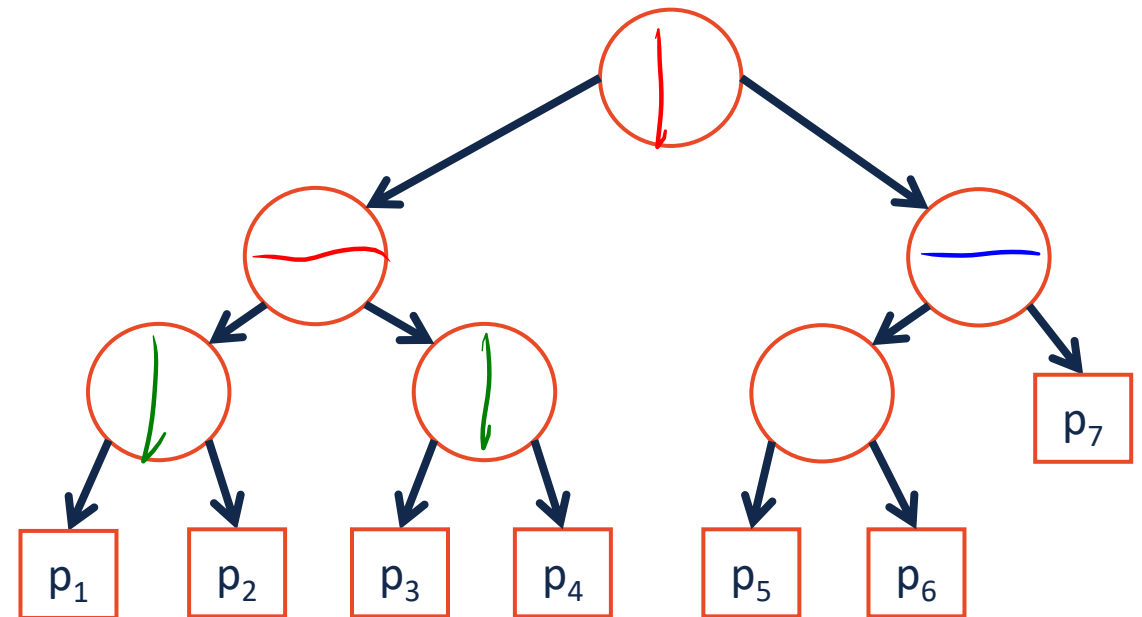
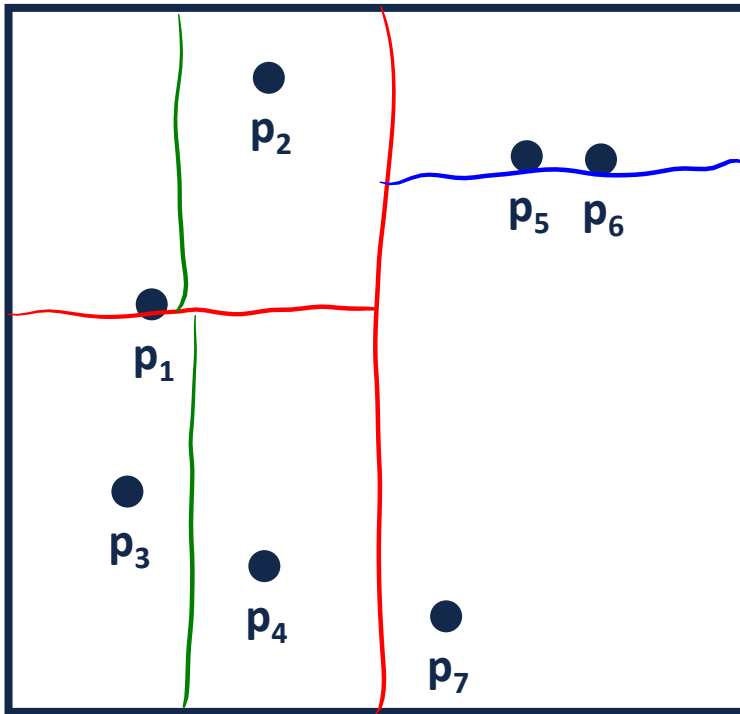
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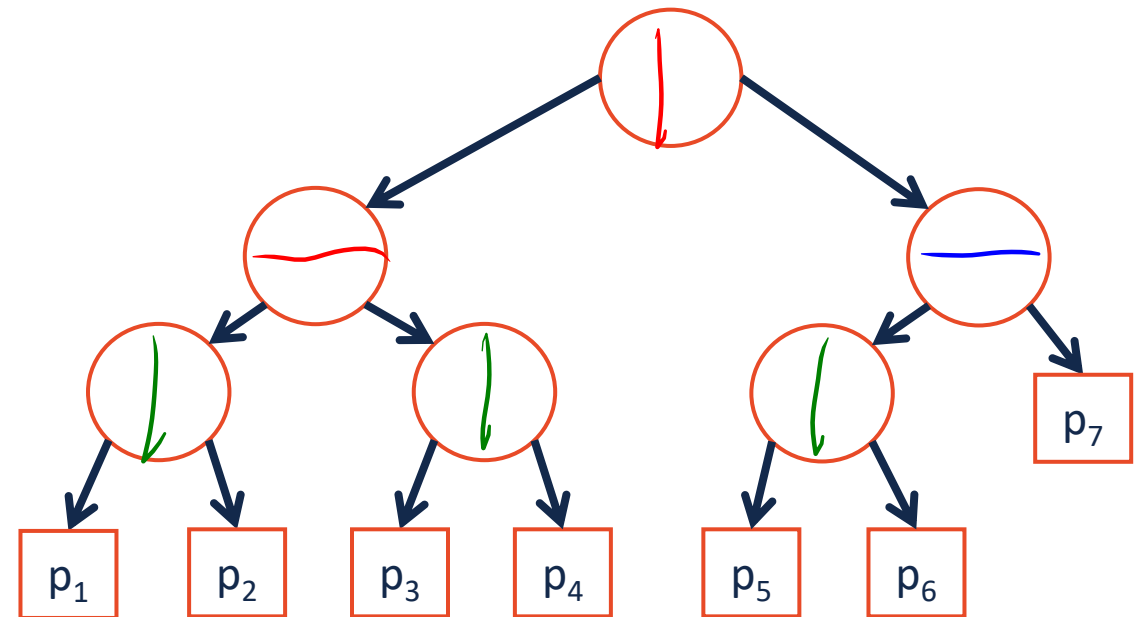
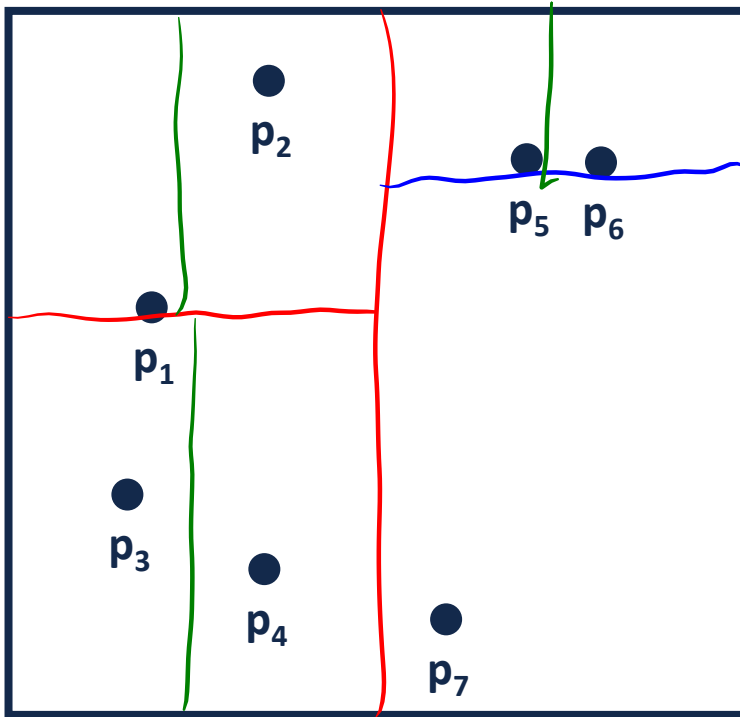
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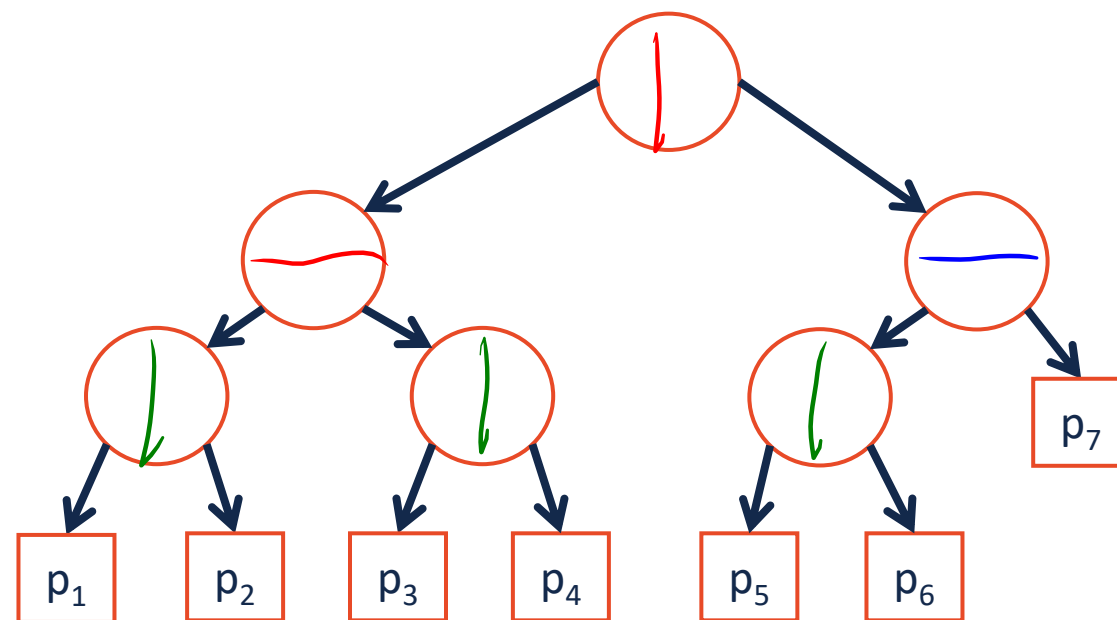
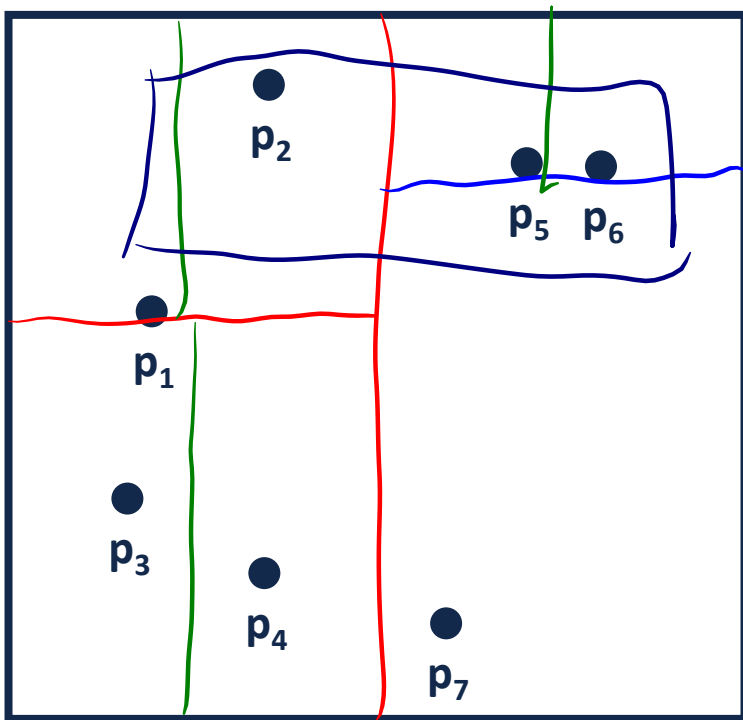
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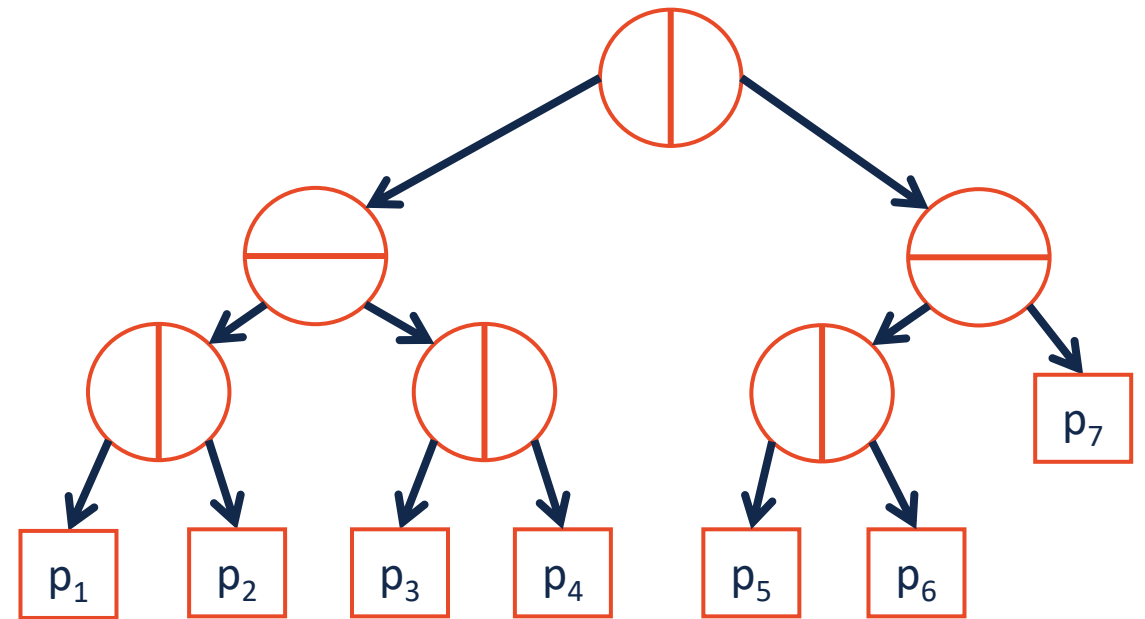
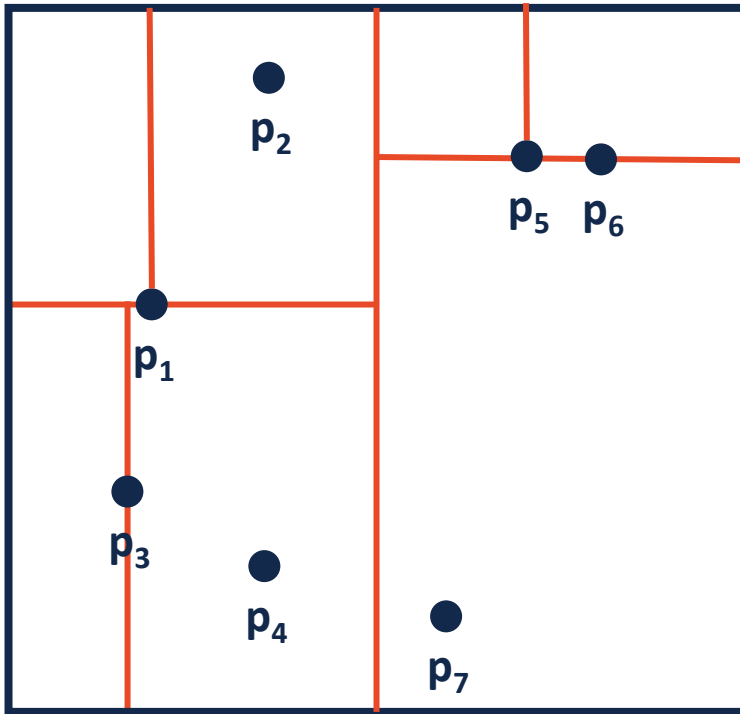
Range-based Searches



Range-based Searches



kD-Trees



CS 225

Data Structures

Volodymyr Kindratenko

Slides courtesy of Wade Fagen-Ulmschneider

BTrees

Q: Can we always fit our data in main memory?

Q: Where else can we keep our data?

However, big-O assumes uniform time for all operations.

BTrees

Q: Can we always fit our data in main memory?

No

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BTrees

Q: Can we always fit our data in main memory?

No

Q: Where else can we keep our data?

Disk

However, big-O assumes uniform time for all operations.

BTrees

Q: Can we always fit our data in main memory?

No

Q: Where else can we keep our data?

Disk

Cloud

However, big-O assumes uniform time for all operations.

Vast Differences in Time

A **3GHz** CPU performs 3M operations in _____.

Old Argument: “Disk Storage is Slow”

- Bleeding-edge storage is pretty fast:

NVMe (M.2, PCIe 3.0 x4):

- Large Disks (10 TB+) still have slow throughput:

New Argument: “The Cloud is Slow!”

Vast Differences in Time

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Old Argument: “Disk Storage is Slow”

- Bleeding-edge storage is pretty fast:

NVMe (M.2, PCIe 3.0 x4):

Read 3500 MBps
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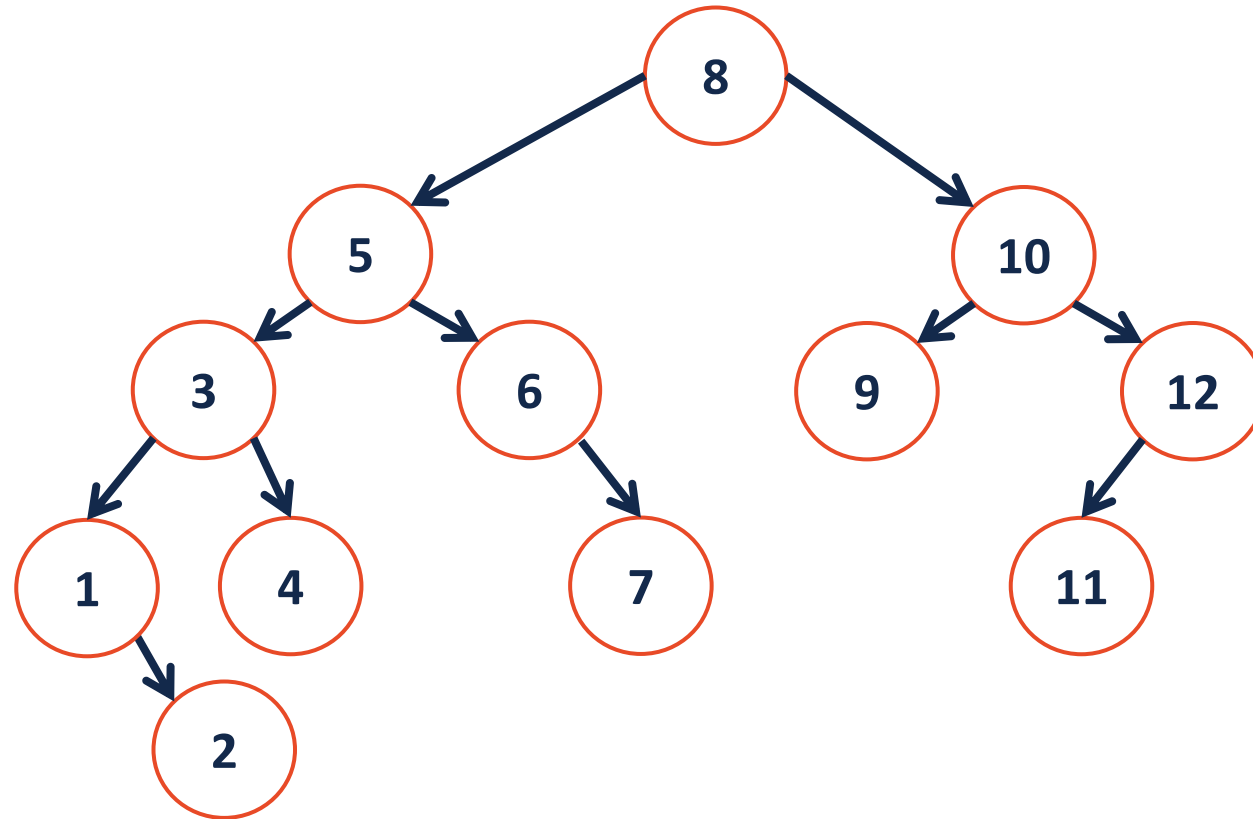
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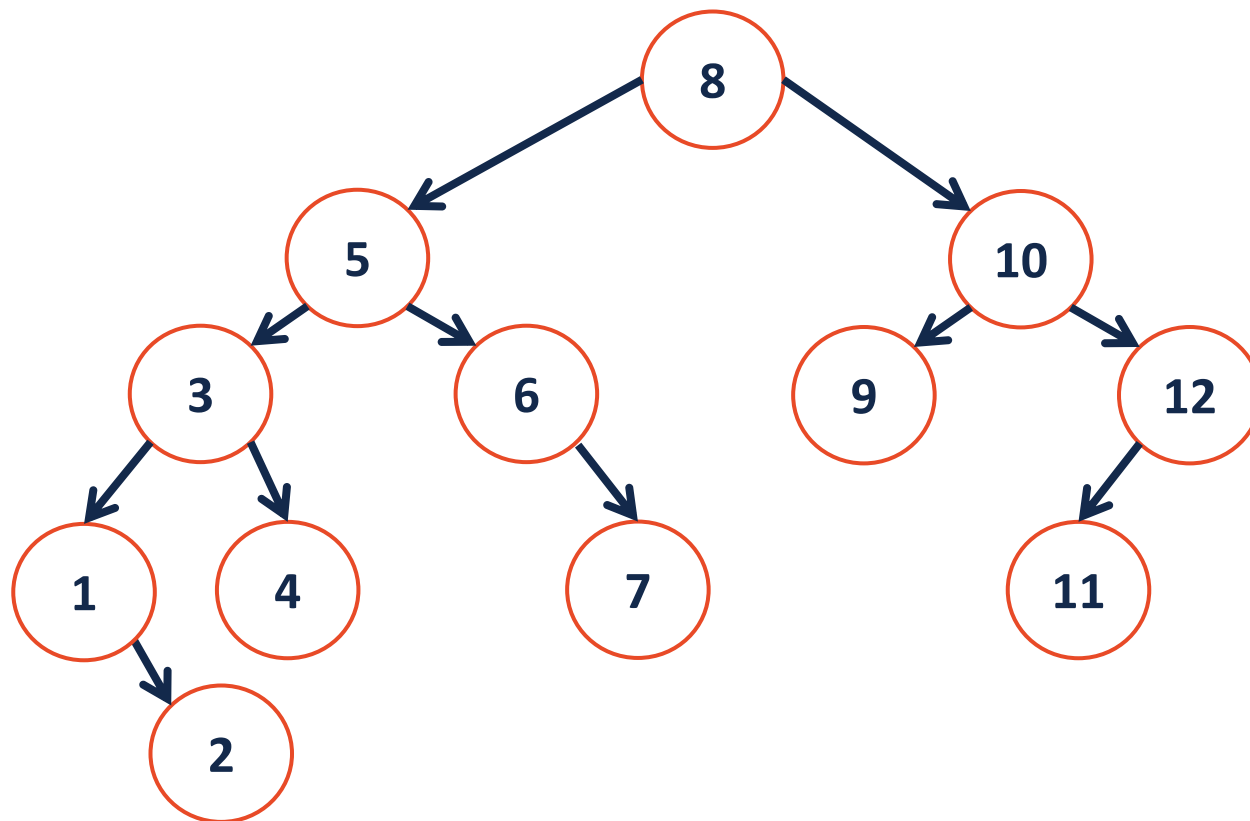
latency 100ms

AVLs on Disk



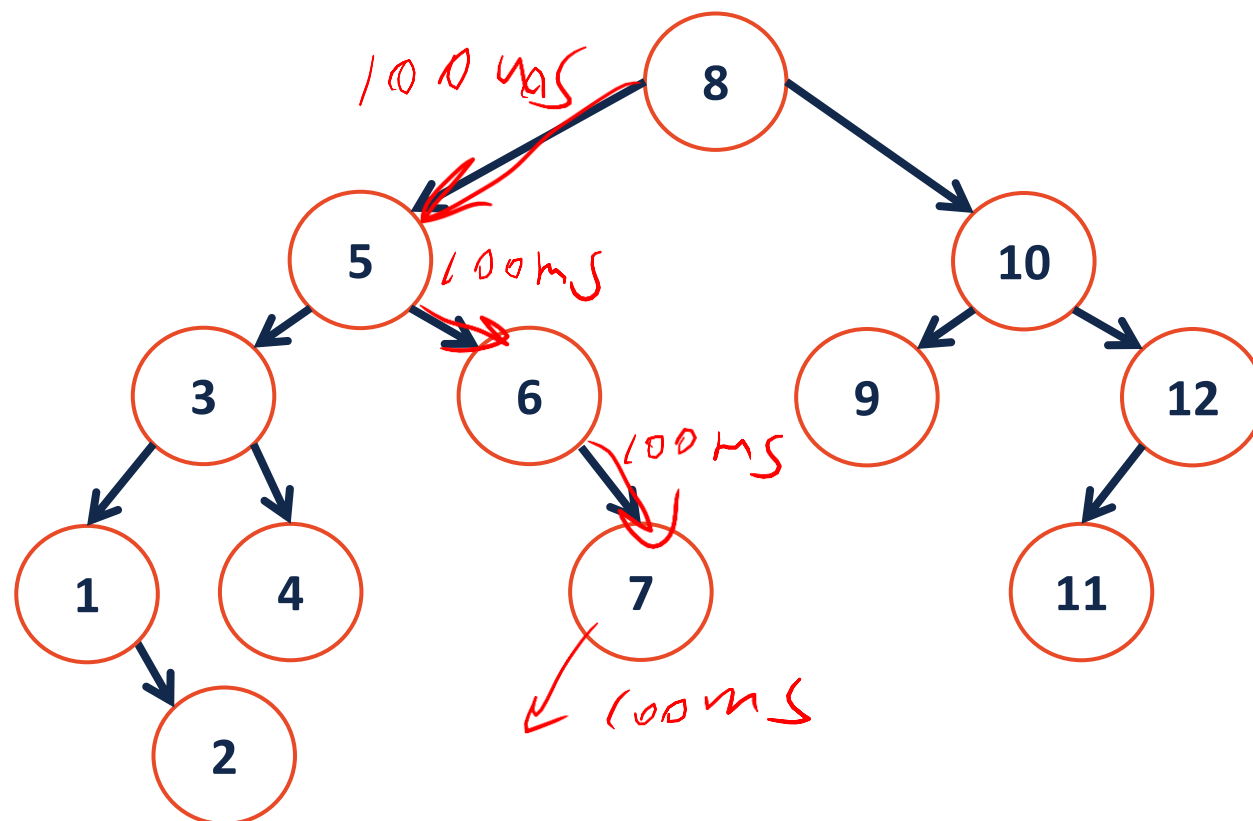
AVLs on Disk

_find(6.5)



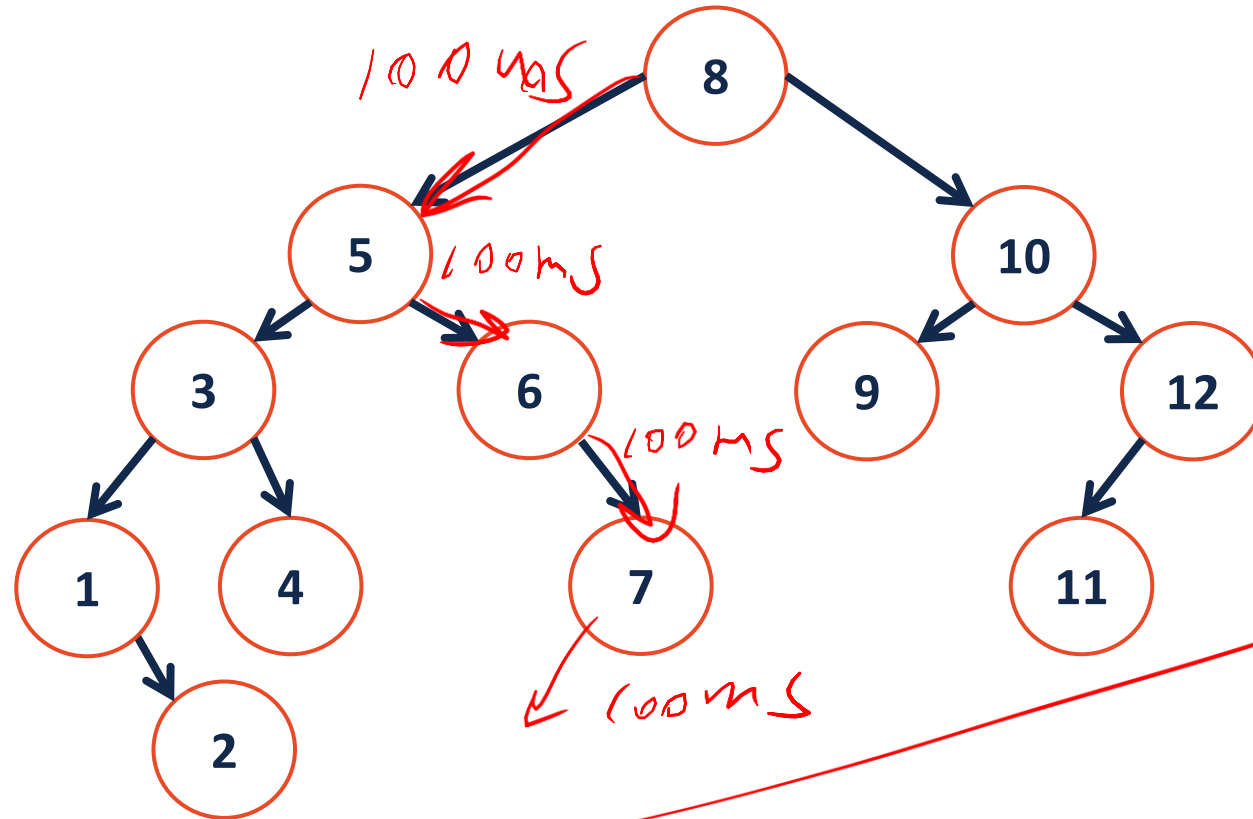
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AVLs on Disk

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400ms Gst w/ n=12

Real Application

Imagine storing driving records for everyone in China:

How many records?

How much data in total?

How deep is the AVL tree?

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↓ $h \approx 30$
3 sec

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BTree Motivations

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Knowing that we have large seek times for data, we want to:

1. keep trees short
2. Make sure data is relevant

BTree (of order m)

-3	8	23	25	31	42	43	55
----	---	----	----	----	----	----	----

m=9

Goal: Minimize the number of reads!

Build a tree that uses _____ / node
[1 network packet]
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BTree Insertion

A **BTree**s of order **m** is an m-way tree:

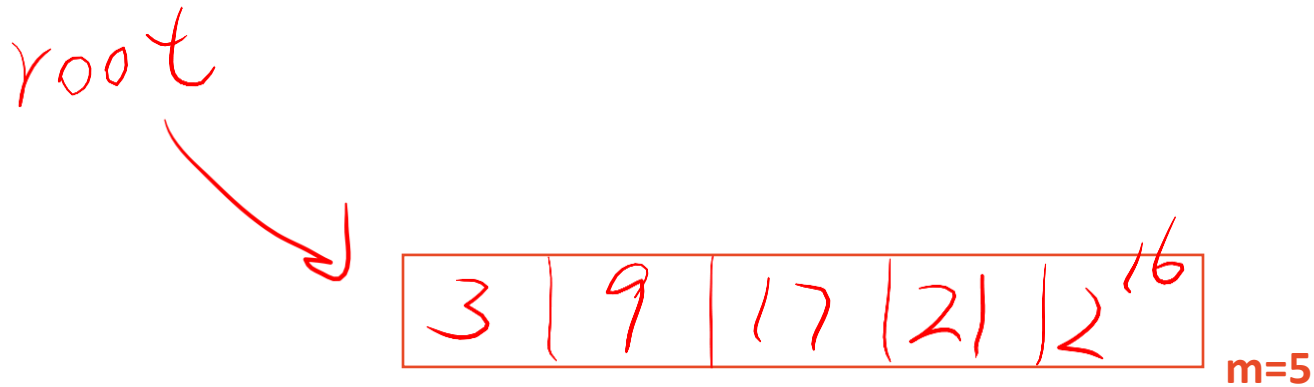
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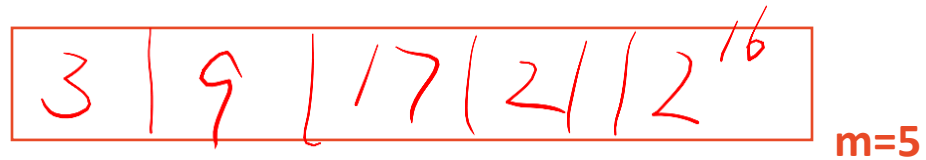
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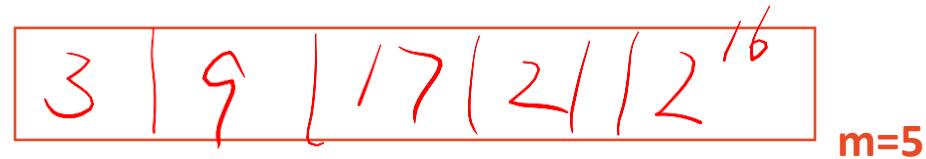
When a BTree node reaches m keys



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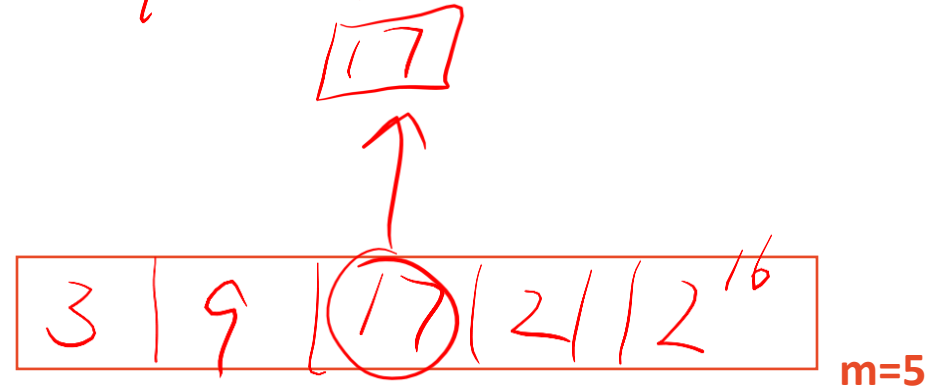
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- "Throw up" the middle node



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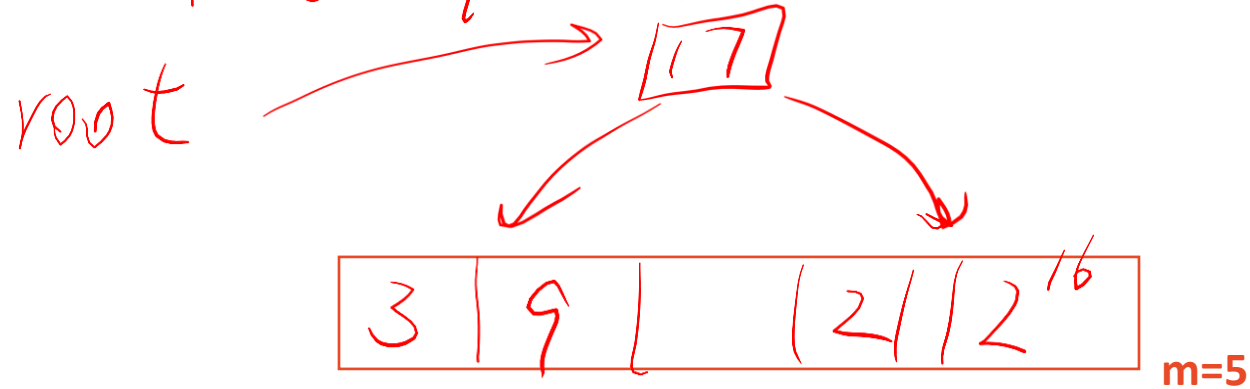
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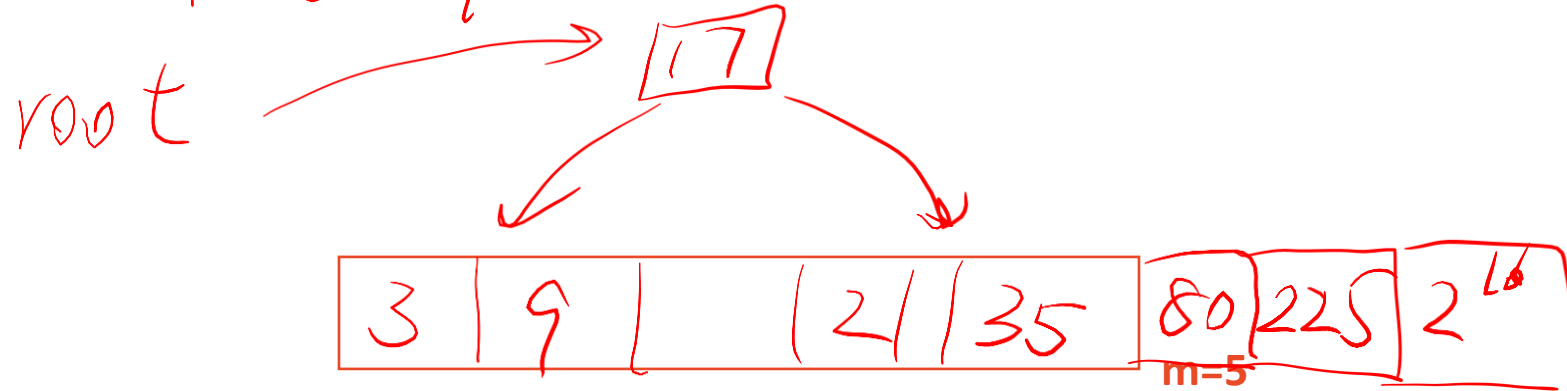
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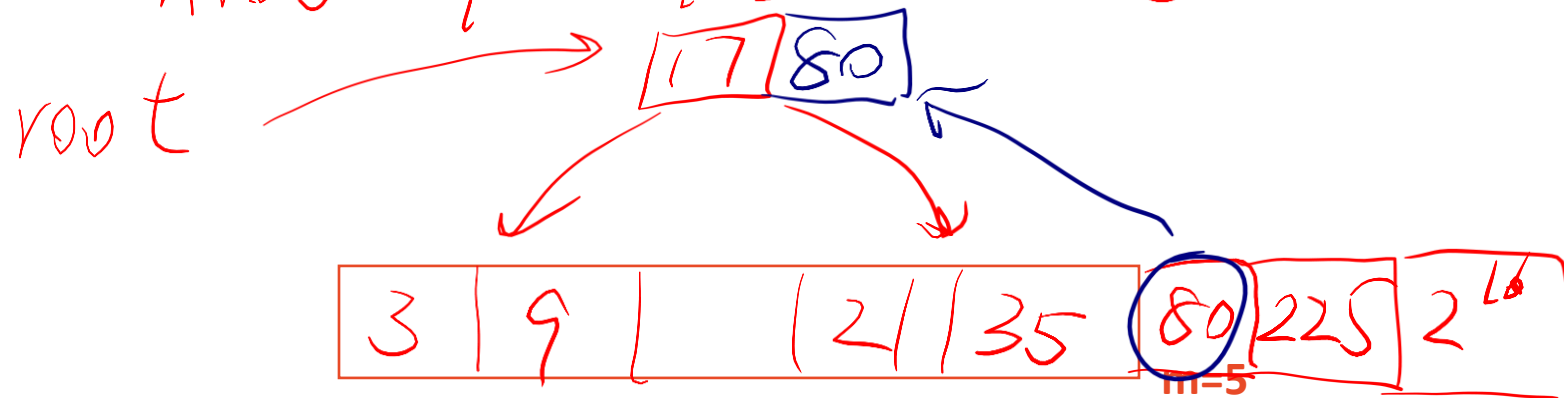
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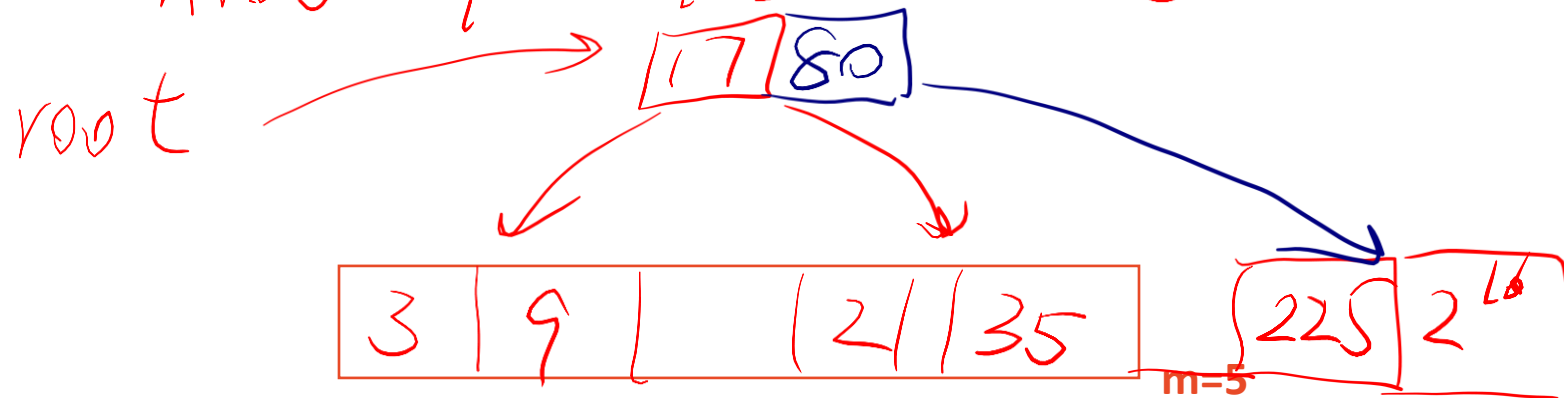
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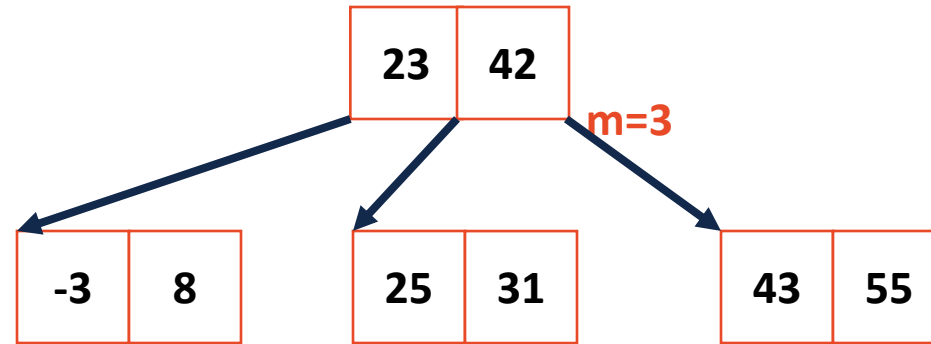
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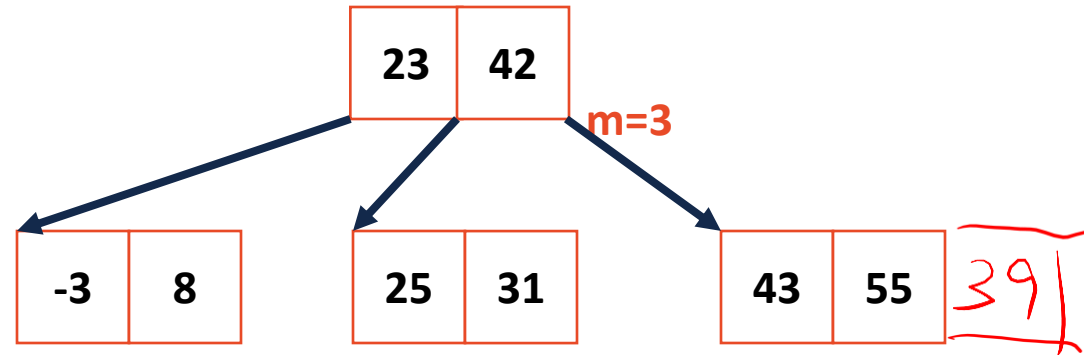
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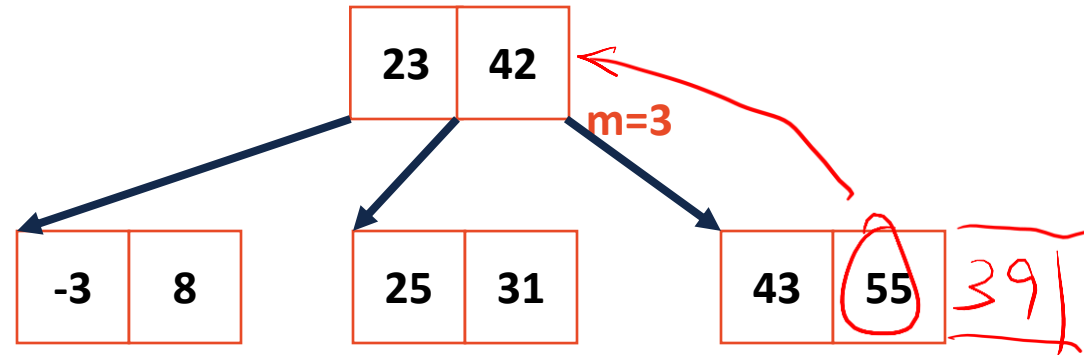
BTree Recursive Insert



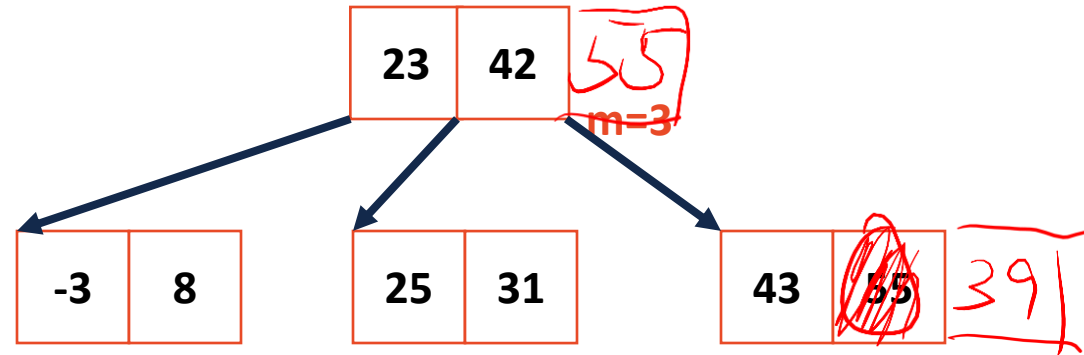
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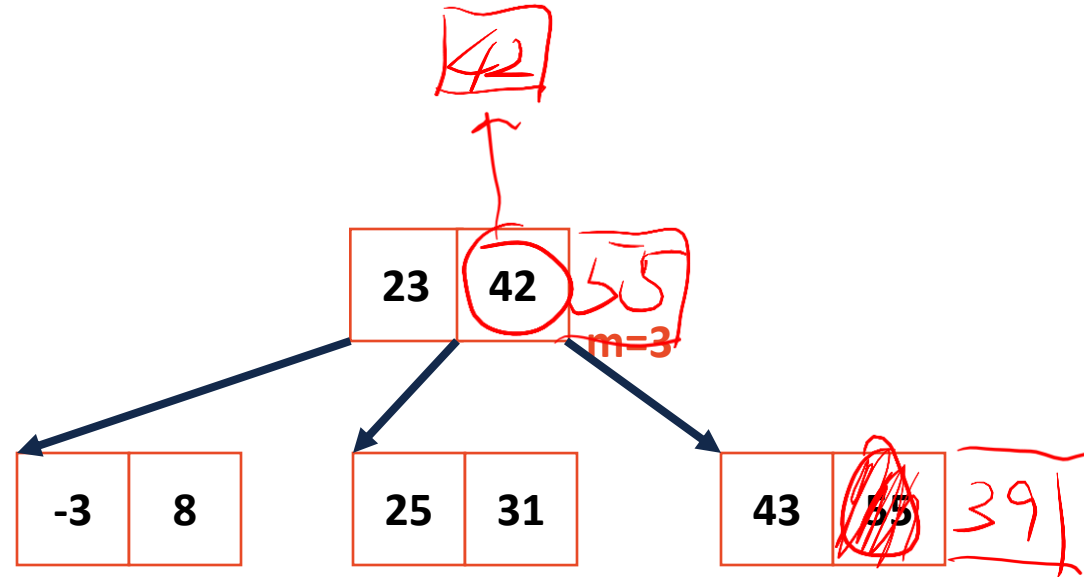
BTree Recursive Insert



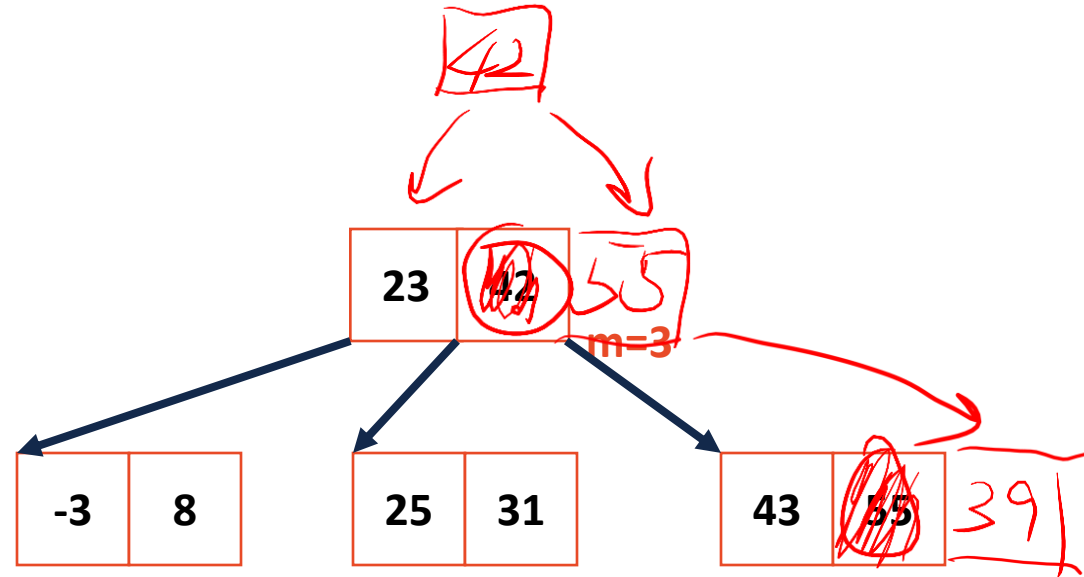
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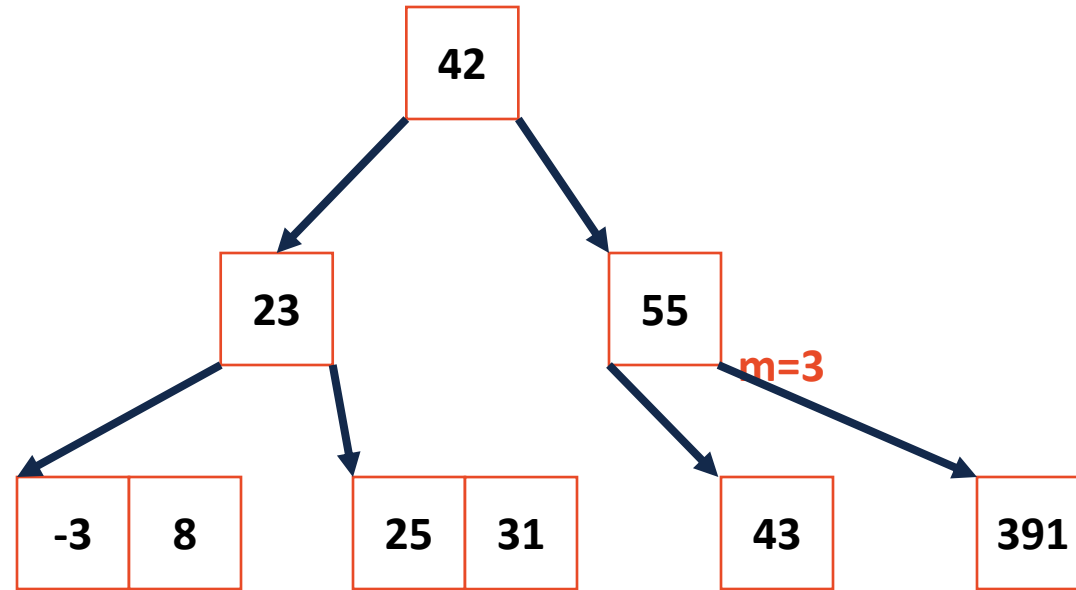
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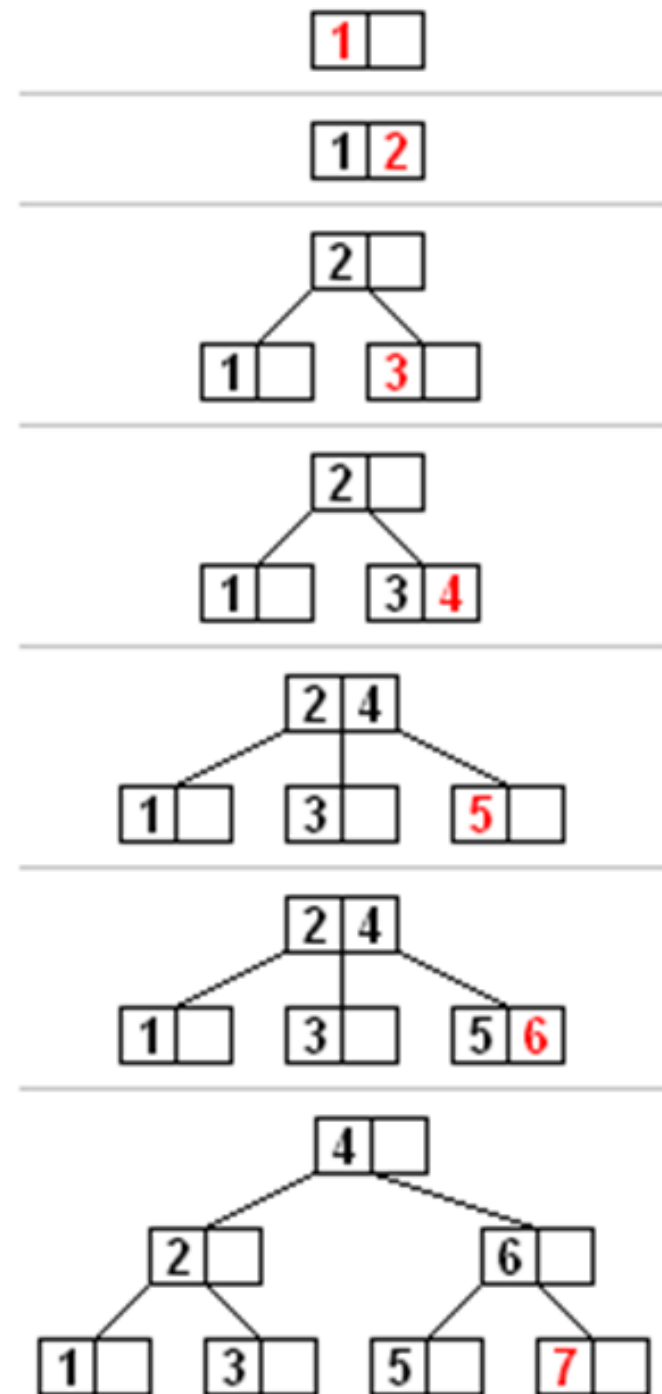
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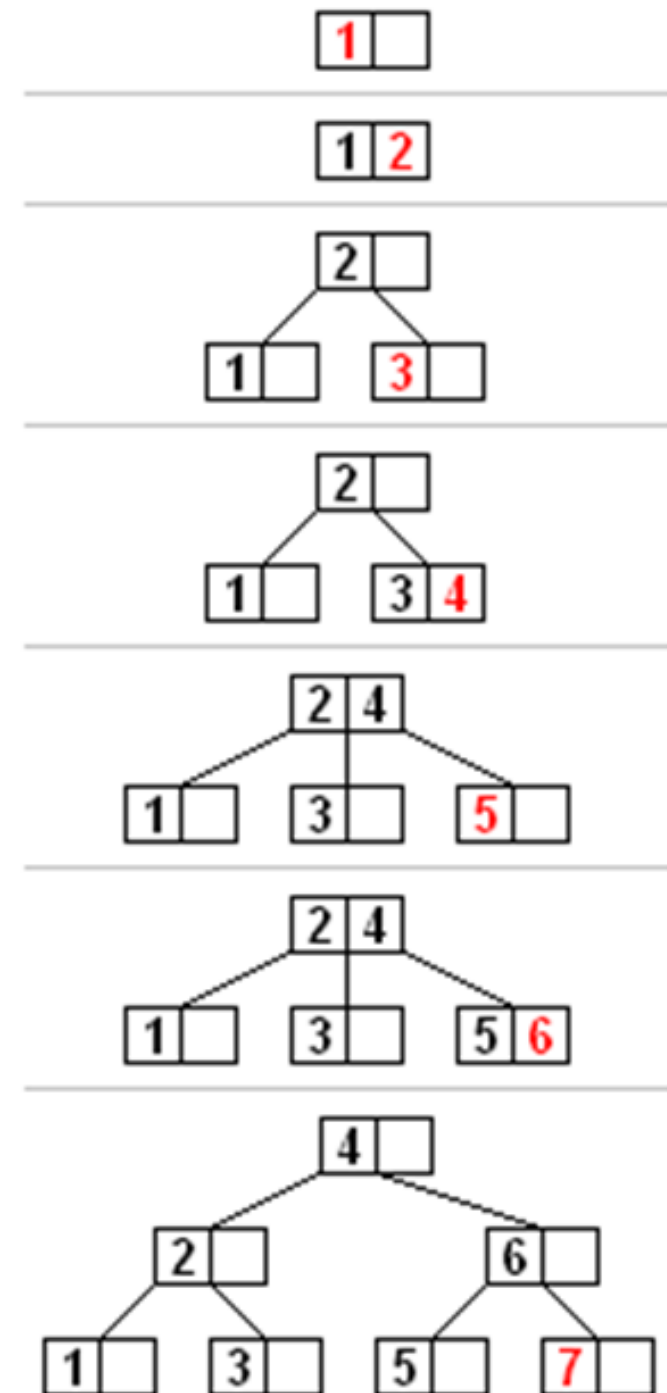


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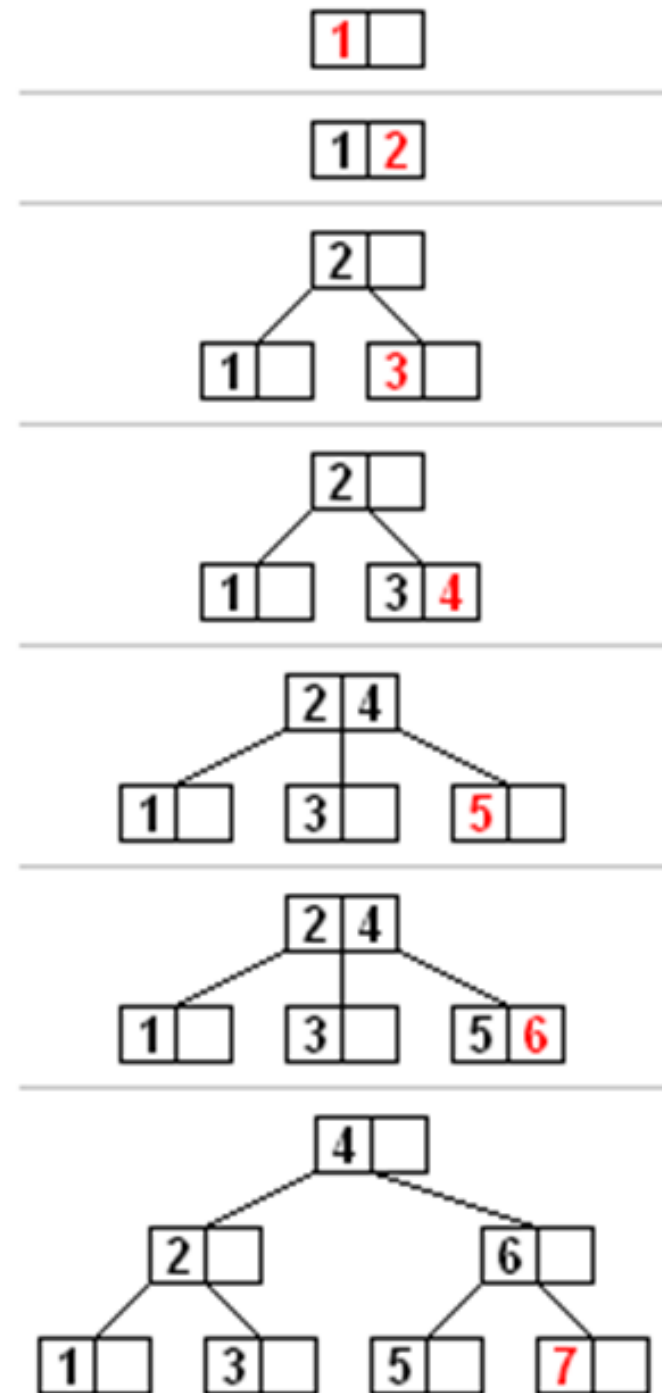
$m = \cancel{2} 3$



BTree Recursive Insert

$$m = \cancel{2} 3$$

All insertions start
at a leaf node



BTree Visualization/Tool

<https://www.cs.usfca.edu/~galles/visualization/BTree.html>

BTree Properties

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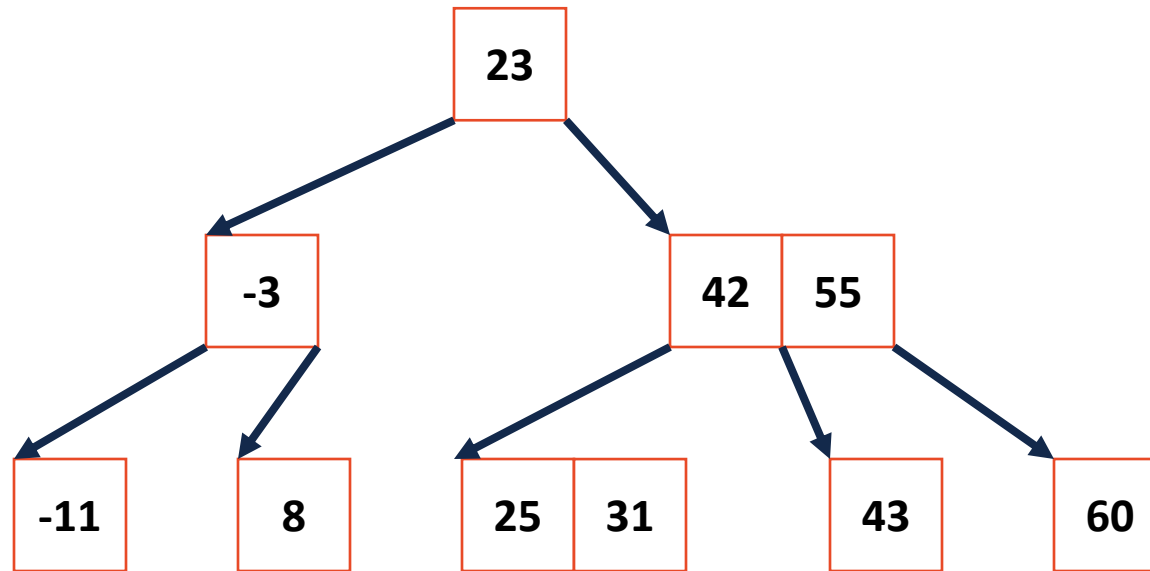
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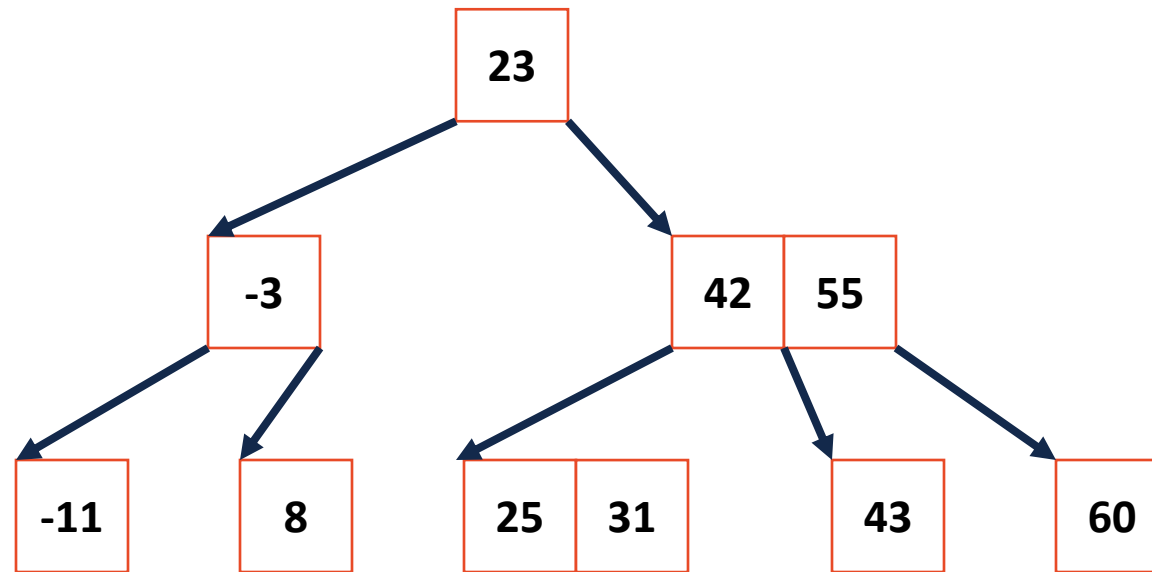
BTree Search



BTree Search

- find (43)

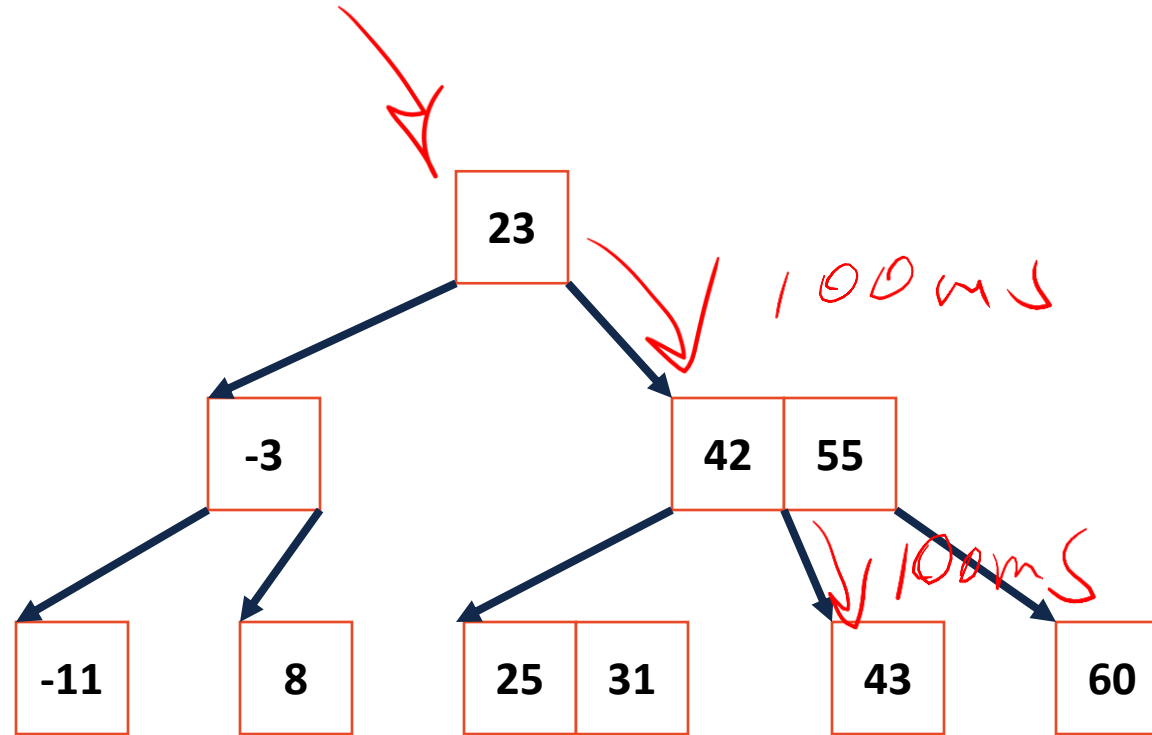
$m=3$



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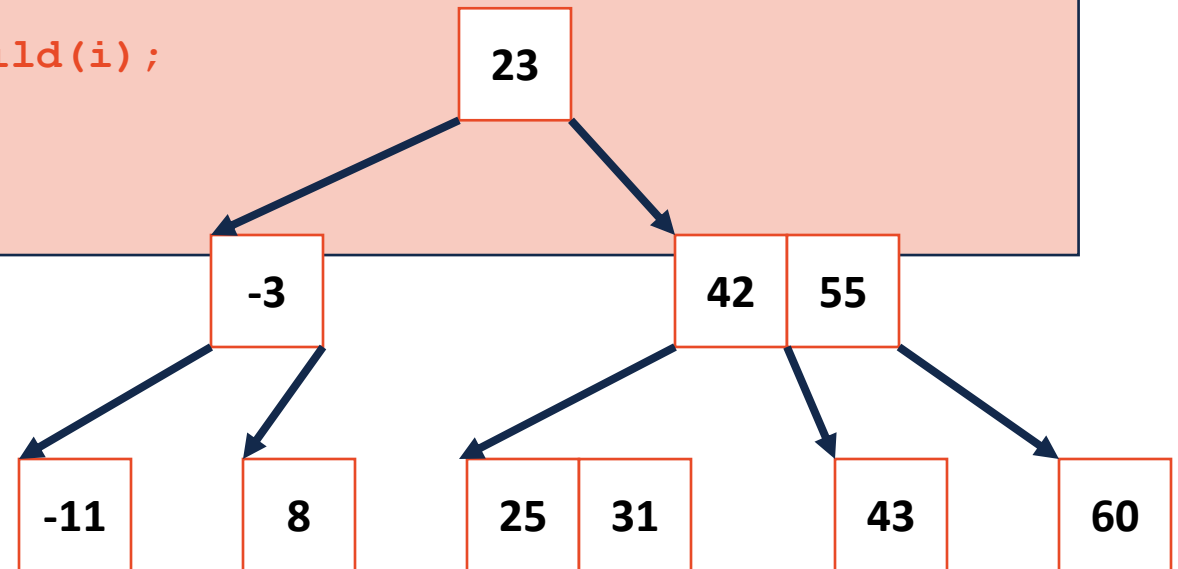
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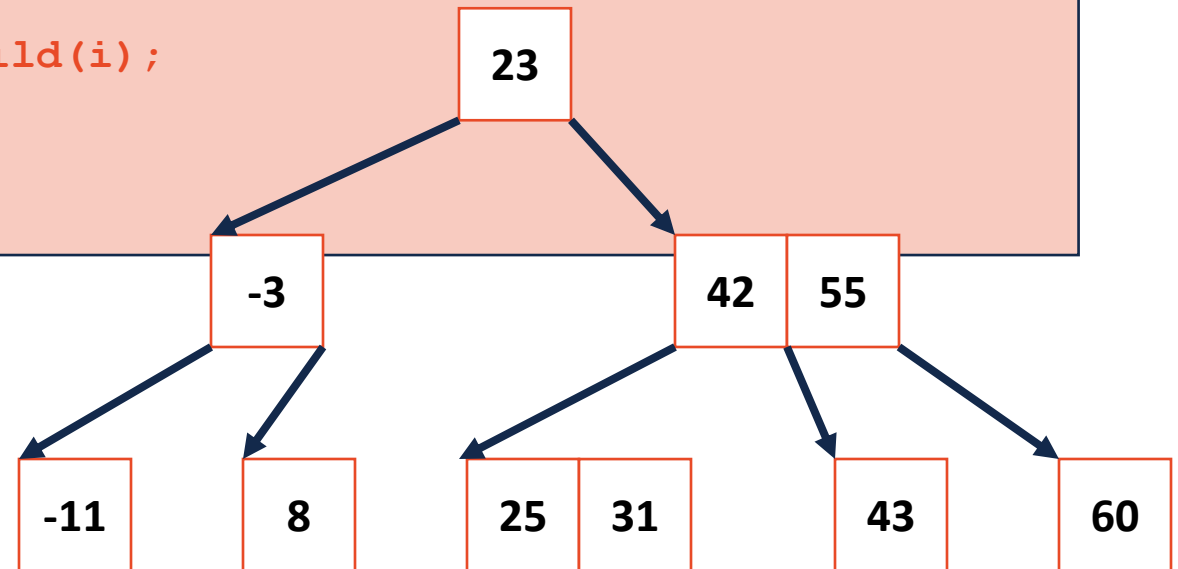
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3     unsigned i;
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6     if ( i < node.keys_ct_ && key == node.keys_[i] ) {
7         return true;
8     }
9
10    if ( node.isLeaf() ) {
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12    } else {
13        BTreeNode nextChild = node._fetchChild(i);
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16 }
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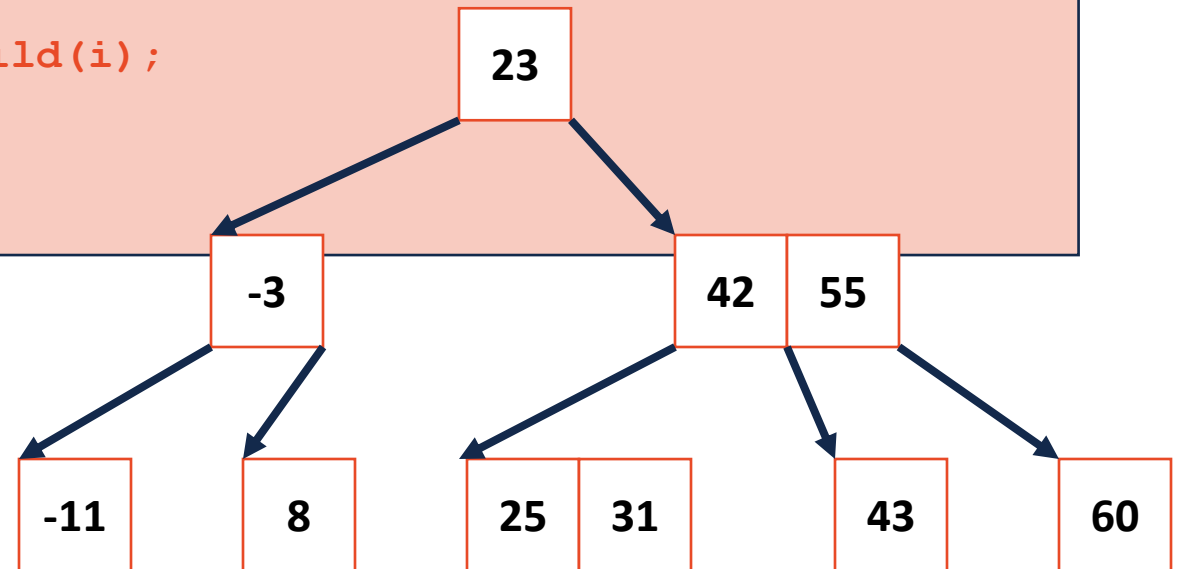
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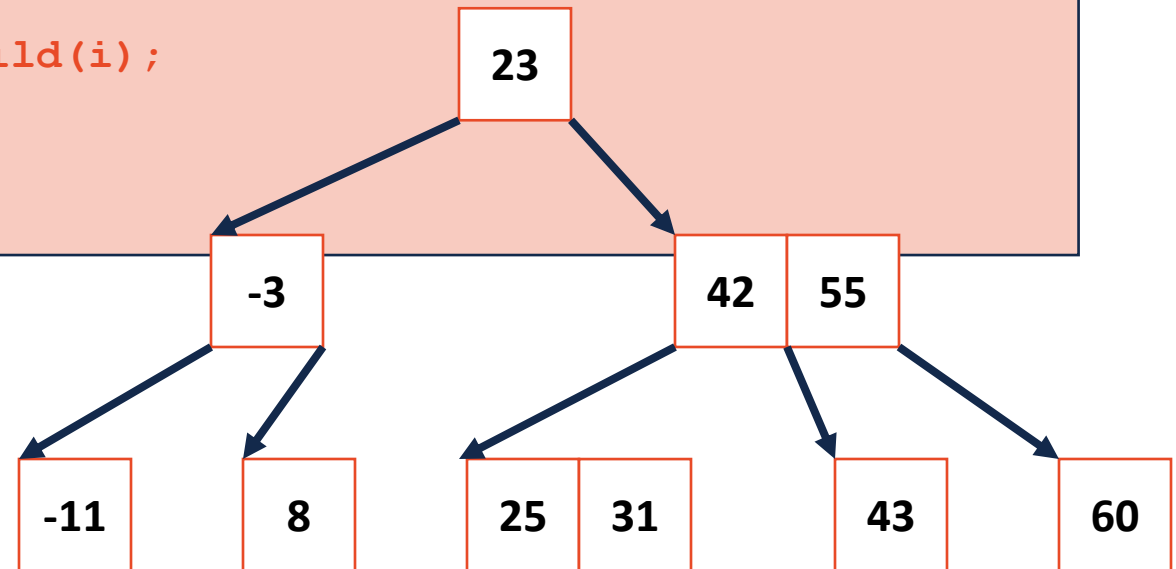
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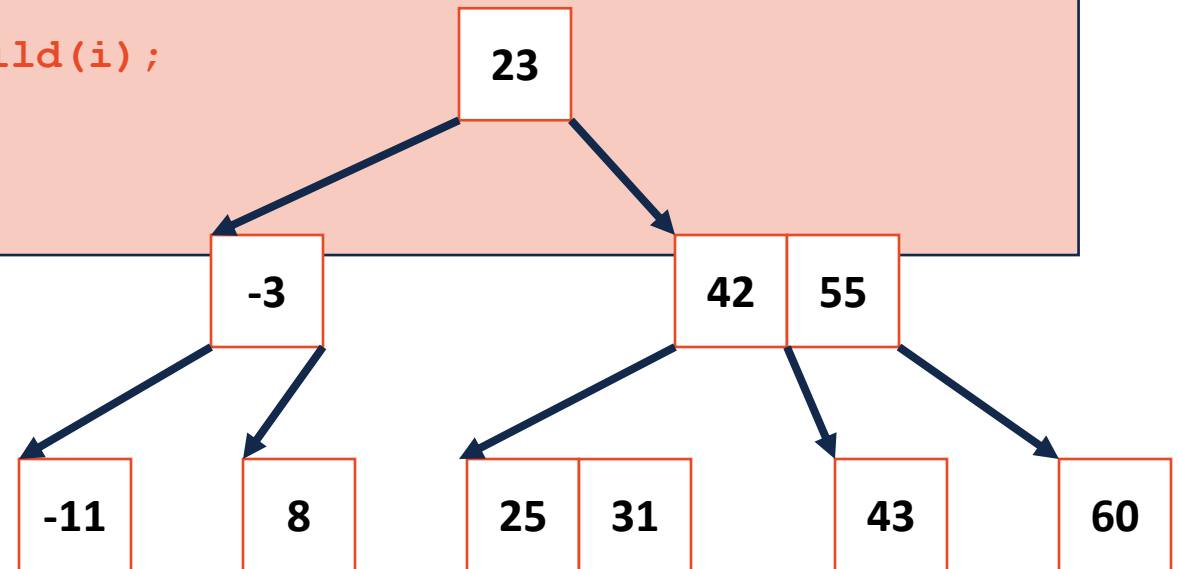


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→ Can do BSearch

→ Recursion

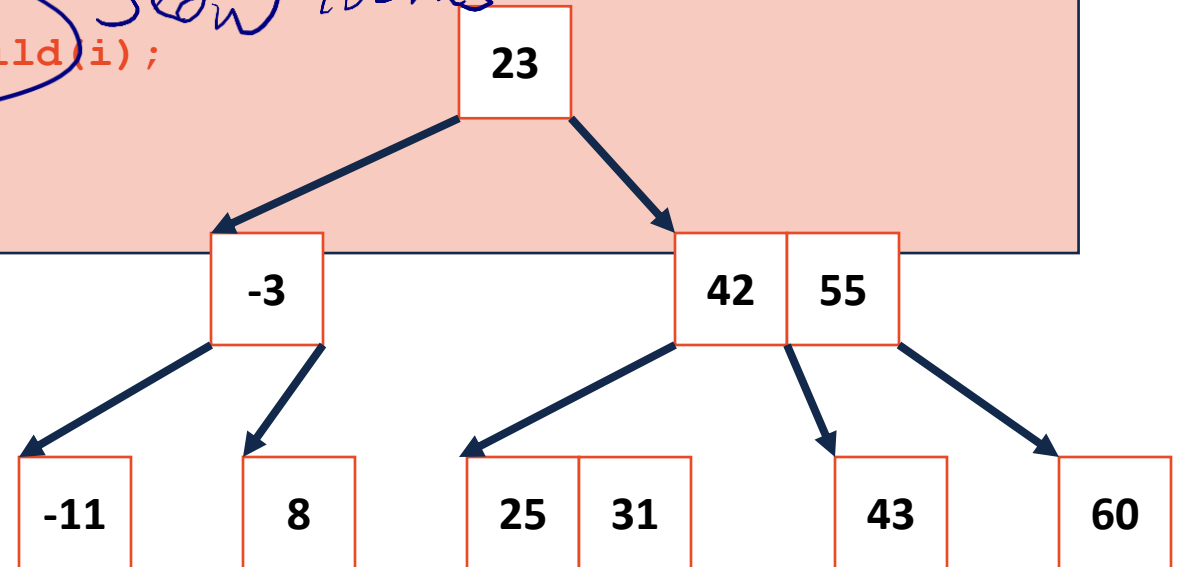


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13        BTreeNode nextChild = node._fetchChild(i); Slow loops  
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→ Can do BSearch

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BTree Analysis

The height of the BTree determines maximum number of _____ possible in data search.

...and the height of the structure is: _____.

Therefore: The number of seeks is no more than _____.

...suppose we want to prove this!

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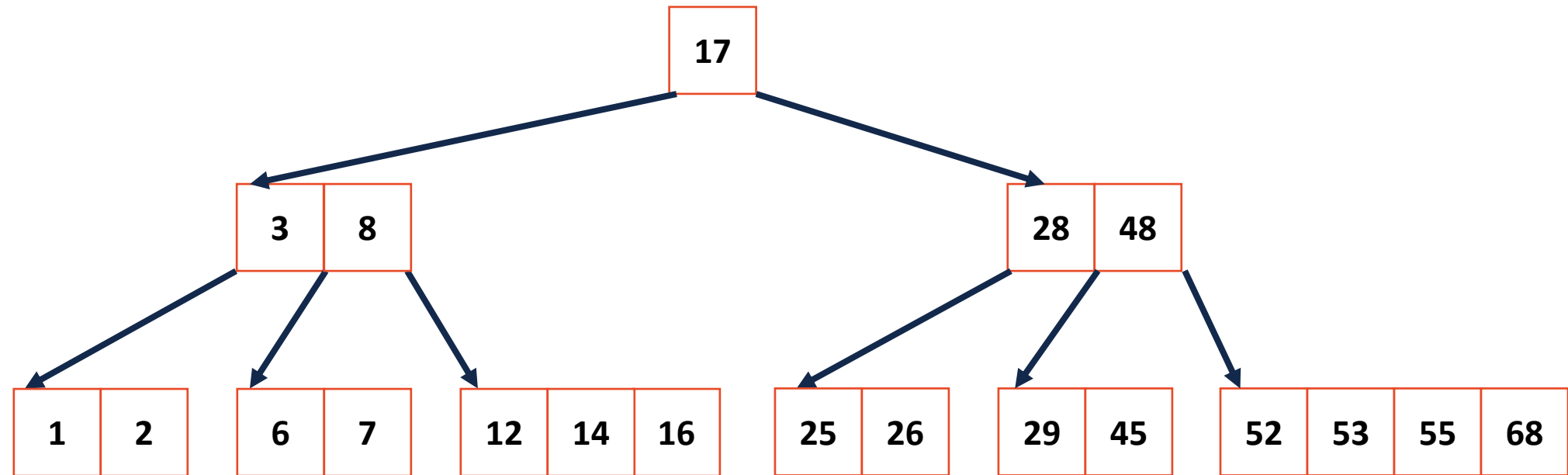
CS 225

Data Structures

Volodymyr Kindratenko

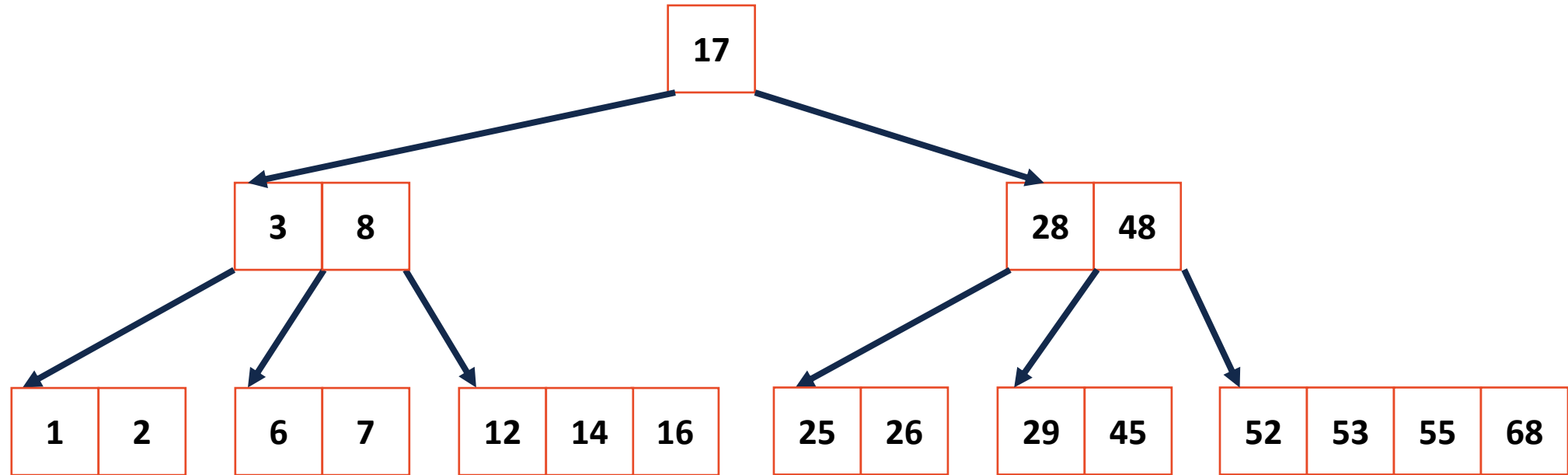
Slides courtesy of Wade Fagen-Ulmschneider

BTree



BTree

order?



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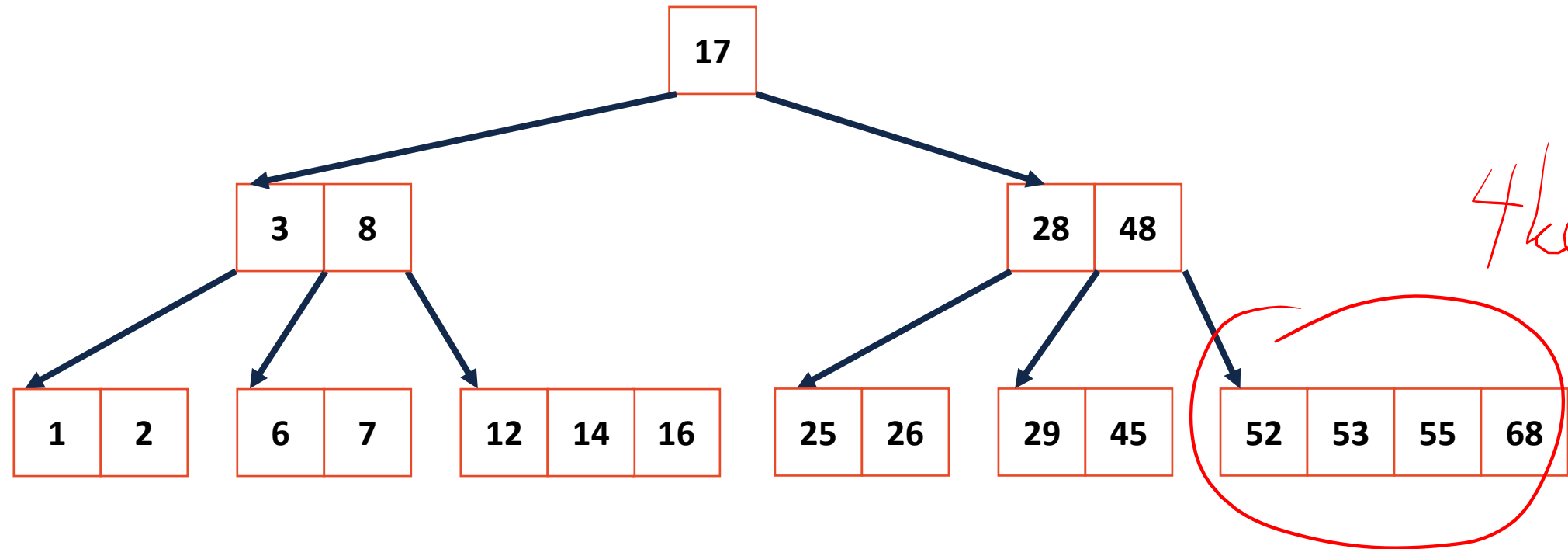
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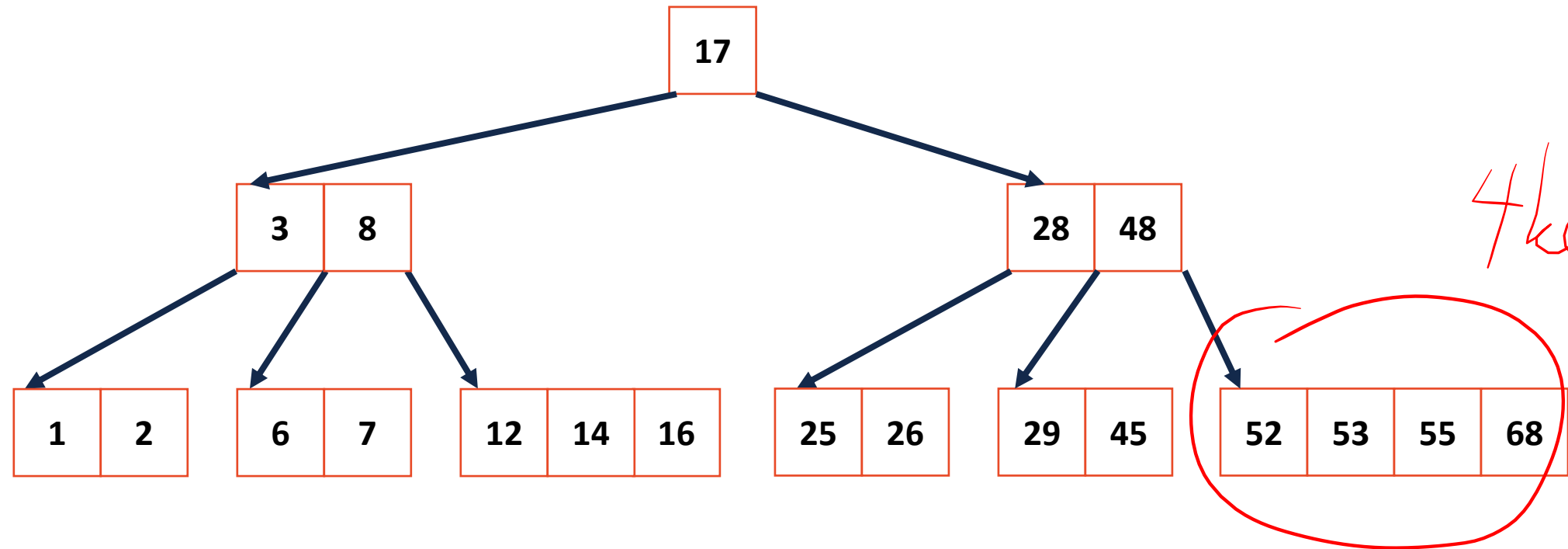
order?



4 keys

BTree

order ? > 4



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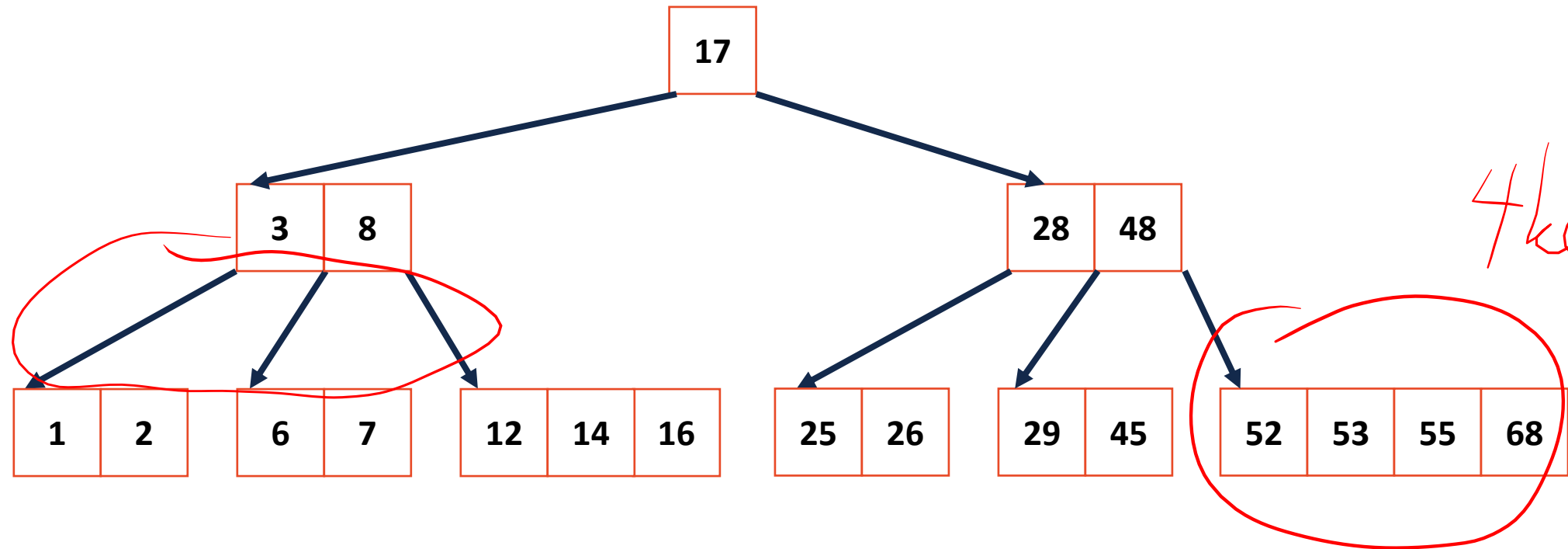
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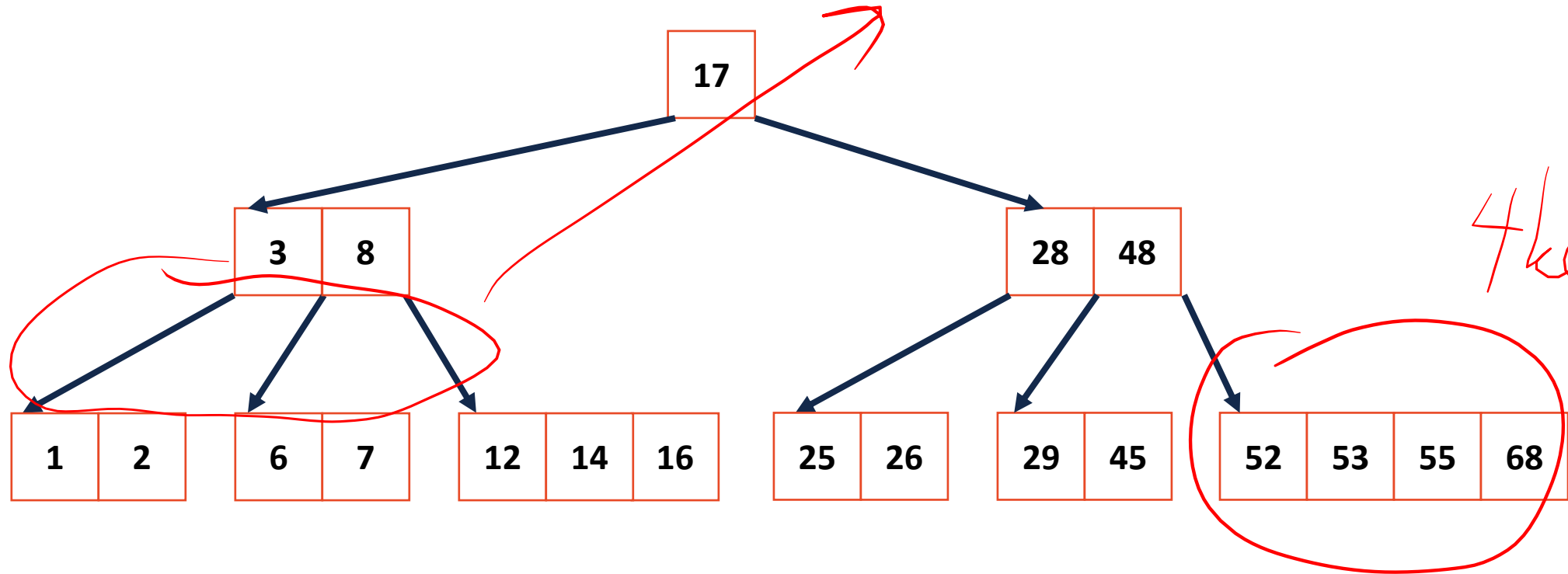
order ? > 4



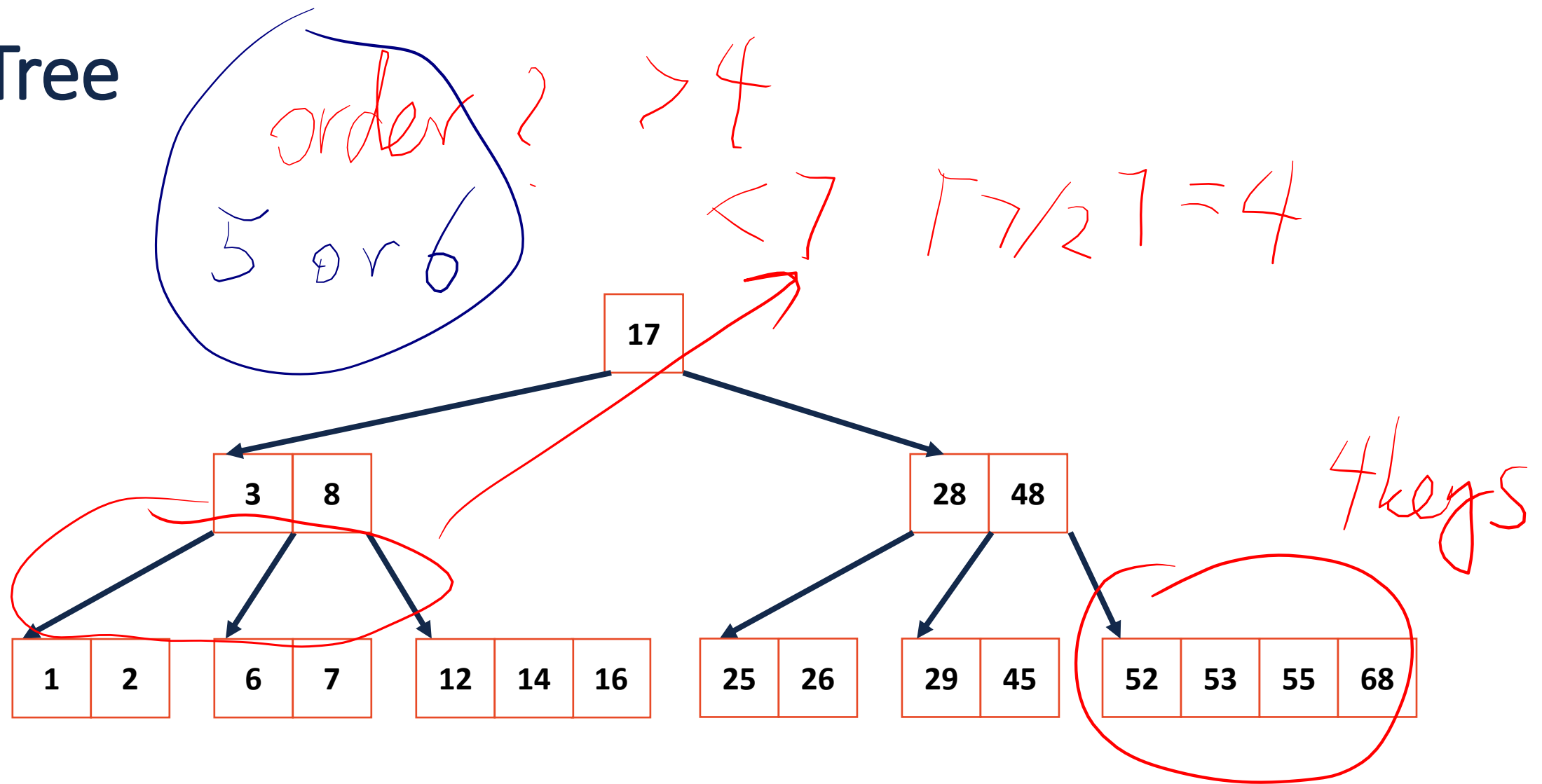
BTree

order ? > 4

< 7 $\lceil 7/2 \rceil = 4$



BTree



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$O(\log n)$?
Therefore: The number of seeks is no more than $\log_m n$.

...suppose we want to prove this!

BTree Analysis

In our AVL Analysis, we saw finding an upper bound on the height (given n) is the same as finding a lower bound on the nodes (given h).

We want to find a relationship for BTrees between the number of keys (n) and the height (h).

BTree Analysis

In our AVL Analysis, we saw finding an upper bound on the height (given **n**) is the same as finding a lower bound on the nodes (given **h**).

$$h \approx f(n)$$

We want to find a relationship for BTrees between the number of keys (**n**) and the height (**h**).

BTree Analysis

In our AVL Analysis, we saw finding an upper bound on the height (given n) is the same as finding a lower bound on the nodes (given h).

$$h \leq f(n)$$

We want to find a relationship for BTrees between the number of keys (n) and the height (h).

$$f(h) \leq n$$

BTree Analysis

Strategy:

We will first count the number of nodes, level by level.

Then, we will add the minimum number of keys per node (**n**).

The minimum number of nodes will tell us the largest possible height (**h**), allowing us to find an upper-bound on height.

BTree Analysis

The minimum number of **nodes** for a BTree of order m **at each level:**

root:

level 1:

level 2:

level 3:

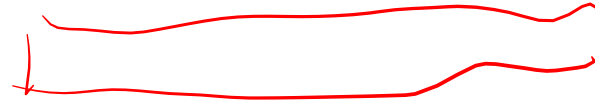
...

level h :

BTree Analysis

The minimum number of **nodes** for a BTree of order m **at each level:**

root:



level 1:

level 2:

level 3:

...

level h :

Btree Properties

A **BTree** of order **m** is an m-way tree:

- All keys within a node are ordered
- All nodes contain no more than **m-1** keys.
- All internal nodes have exactly **one more child than keys**
- Root nodes can be a leaf or have **[2, m]** children.
- All non-root, internal nodes have **[ceil(m/2), m]** children.
- All leaves are on the same level

BTree Analysis

The minimum number of **nodes** for a BTree of order m **at each level:**

root:

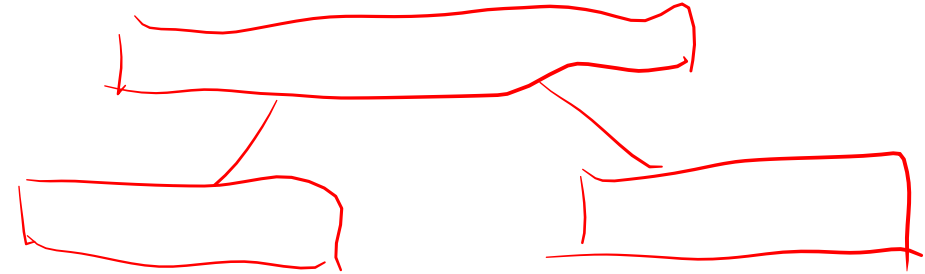
level 1:

level 2:

level 3:

...

level h :



BTree Analysis

$$\text{Let } t = \lceil m/2 \rceil$$

The minimum number of **nodes** for a BTree of order m **at each level:**

root:

1

level 1:

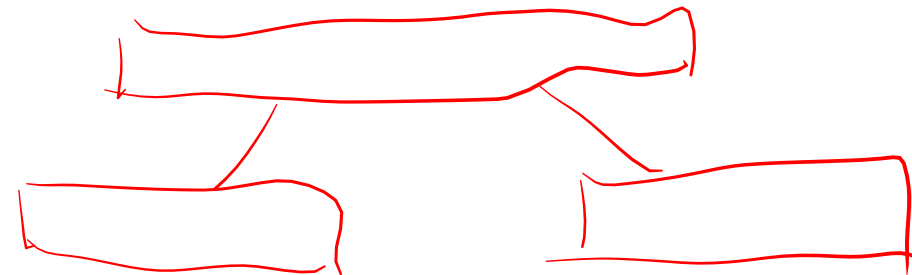
2

level 2:

level 3:

...

level h :



BTree Analysis

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The minimum number of **nodes** for a BTree of order m **at each level**:

root:

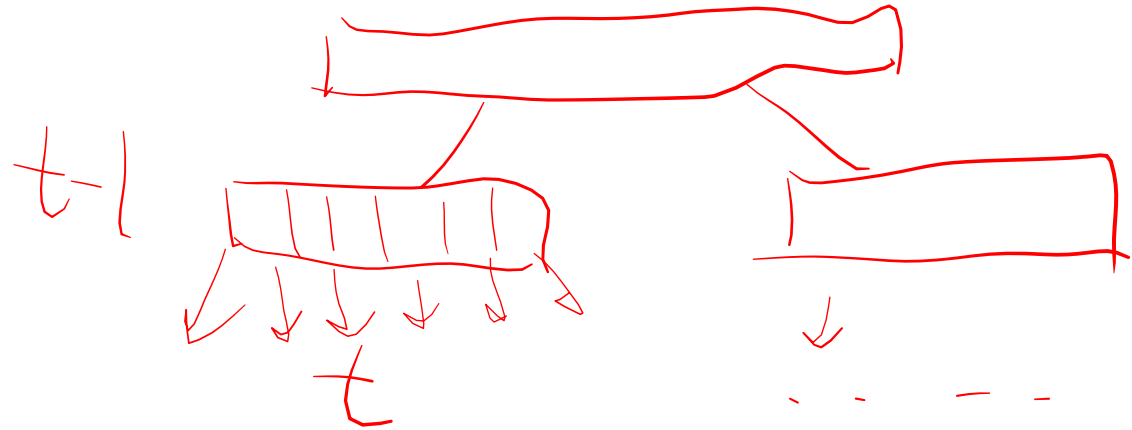
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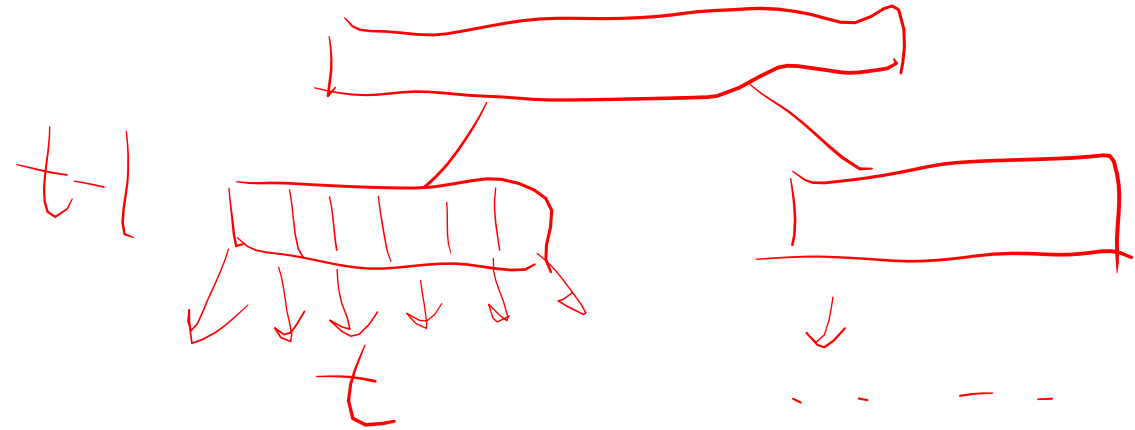
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level 2:

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...

level h :



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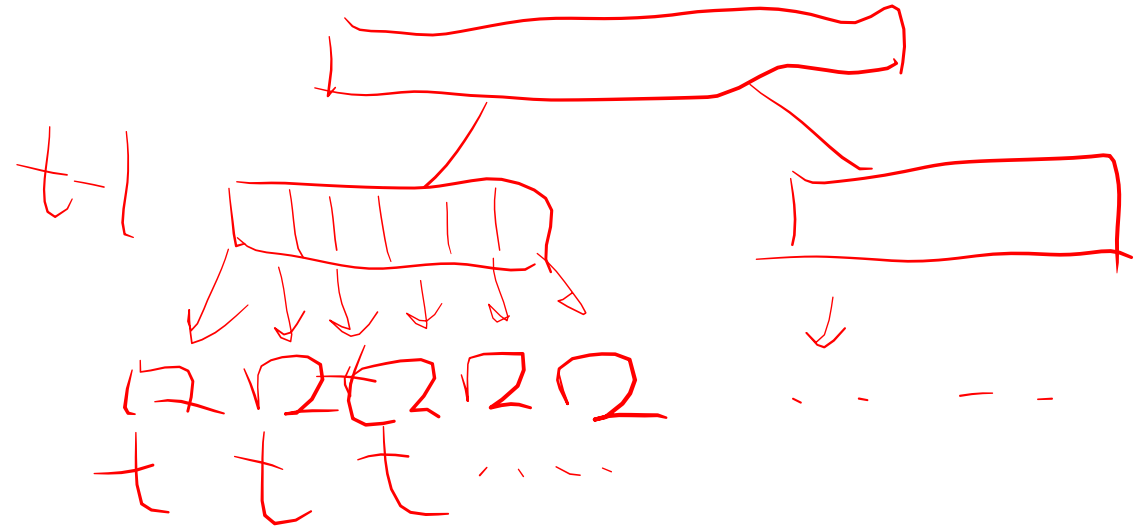
level 2:

level 3:

...

level h :

1
2
2t



BTree Analysis

$$\text{Let } t = \lceil m/2 \rceil$$

The minimum number of **nodes** for a BTree of order m **at each level**:

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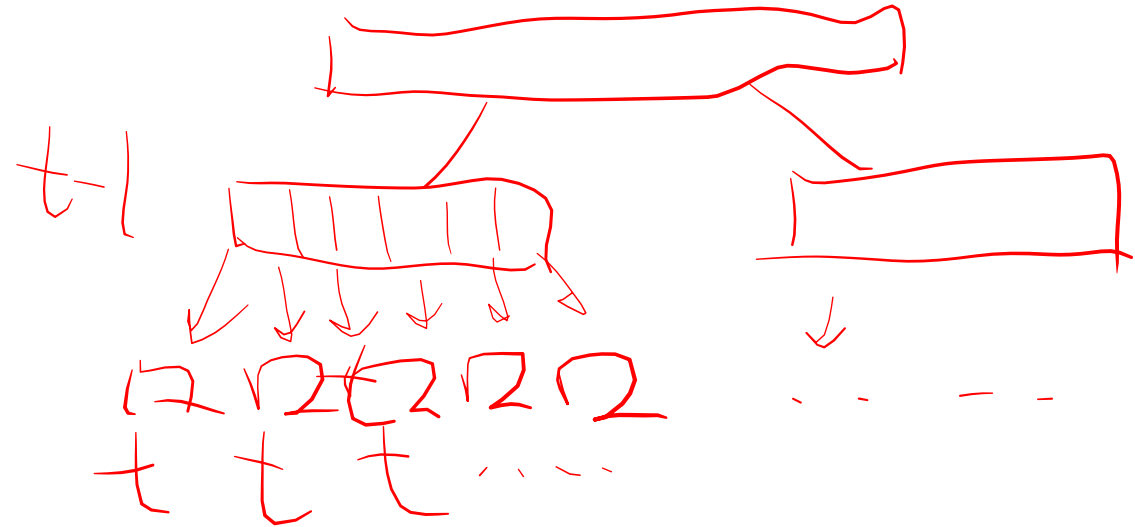
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BTree Analysis

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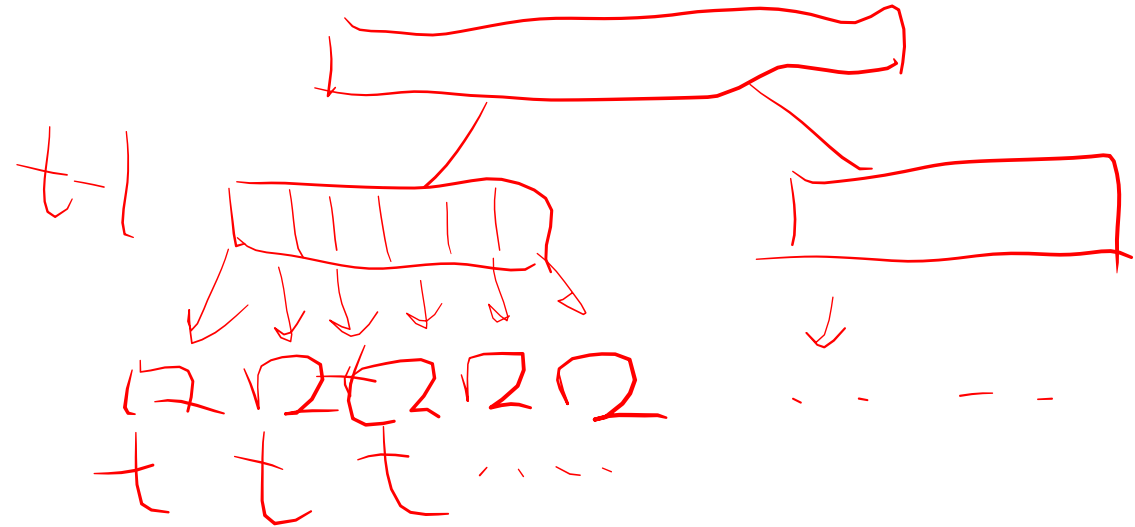
1

2

$2t$

$2t^2$

$2t^{h-1}$



BTree Analysis

The **total number of nodes** is the sum of all of the levels:

BTree Analysis

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$$= 1 + 2 + 2t + 2t^2 + \dots + 2t^{h-1}$$

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BTree Analysis

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$$= 1 + 2 \cdot \frac{t^{h-1+1} - 1}{t - 1}$$

BTree Analysis

The **total number of nodes** is the sum of all of the levels:

$$\begin{aligned} &= 1 + 2 + 2t + 2t^2 + \dots + 2t^{h-1} \\ &= 1 + 2 \sum_{k=1}^{h-1} t^k \\ &= 1 + 2 \cdot \frac{t^{h-1+1} - 1}{t - 1} = 1 + 2 \left(\frac{t^h - 1}{t - 1} \right) \end{aligned}$$

BTree Analysis

The **total number of nodes** is the sum of all of the levels:

$$= 1 + 2t + 2t^2 + \dots + 2t^{h-1}$$

$$= 1 + 2 \sum_{k=1}^{h-1} t^k$$

$$= 1 + 2 \cdot \frac{t^{h-1+1} - 1}{t - 1} = 1 + 2 \left(\frac{t^h - 1}{t - 1} \right)$$

min number of nodes in Btree of order m

BTree Analysis

The **total number of keys**:

BTree Analysis

The **total number of keys:**

Each non-root node has $t-1$ keys

BTree Analysis

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$$1 + 2 \left(\frac{t^h - 1}{t - 1} \right)$$

BTree Analysis

The **total number of keys**:

Each non-root node has $t-1$ keys

$$1 + 2 \left(\frac{t^h - 1}{t - 1} \right) (t - 1)$$

BTree Analysis

The **total number of keys:**

Each non-root node has $t-1$ keys

$$1 + 2 \left(\frac{t^h - 1}{t - 1} \right) - (\cancel{t-1})$$

BTree Analysis

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Each non-root node has $t-1$ keys

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$$= 2t^h - 1$$

BTree Analysis

The **total number of keys**:

Each non-root node has $t-1$ keys

$$1 + 2 \left(\frac{t^h - 1}{t - 1} \right) - \left(\cancel{t-1} \right)$$

$= 2t^h - 1$ min keys
a B tree of order m
given h

BTree Analysis

Given **m=101**, a tree of height **h=4** has:

Minimum Keys:

Maximum Keys:

BTree Analysis

order of Btree

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order of Btree

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Minimum Keys:

$$2^h - 1 \quad h=4 \rightarrow$$

Maximum Keys:

BTree Analysis

order of Btree

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Minimum Keys:

$$2t^h - 1 \xrightarrow{h=4} 2t^4 - 1$$

Maximum Keys:

BTree Analysis

order of Btree

Given $m=101$, a tree of height $h=4$ has:

Minimum Keys:

$$2t^h - 1 \xrightarrow{h=4} 2t^4 - 1 \xrightarrow{t=\lceil m/2 \rceil = 51}$$

Maximum Keys:

BTree Analysis

order of Btree

Given **m=101**, a tree of height **h=4** has:

Minimum Keys:

$$2t^h - 1 \xrightarrow{h=4} 2t^4 - 1 \xrightarrow{t=\lceil m/2 \rceil = 51} 2(51)^4 - 1 \approx 12.5m$$

Maximum Keys:

BTree Analysis

order of Btree

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Minimum Keys:

$$2t^h - 1 \xrightarrow{h=4} 2t^4 - 1 \xrightarrow{t=\lceil m/2 \rceil = 51}$$

$$2(51)^4 - 1 \approx 12.5m$$

Maximum Keys:

$$\boxed{\text{only } 300m}$$

BTree Analysis

The **smallest total number of keys** is:

So an inequality about **n** , the total number of keys:

Solving for **h** , since **h** is the number of seek operations:

BTree Analysis

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$m = 101$
 $n = 1m$

BTree Analysis

The **smallest total number of keys** is:

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$$\log_t \left(\frac{n+1}{2} \right) \geq h \rightarrow \log_{(m/2)} \left(\frac{n+1}{2} \right) \geq h$$

$$\log_{(5)}(500k) = 3.37$$

$$m = 101$$
$$h = \lceil m \rceil$$

BTree Analysis

order of Btree

Given **m=101**, a tree of height **h=4** has:

Minimum Keys:

$$2t^h - 1 \xrightarrow{h=4} 2t^4 - 1 \xrightarrow{t=\lceil m/2 \rceil = 51}$$

$$2(51)^4 - 1 \approx 12.5m$$

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BTree Analysis

order of Btree

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Prove yourself!

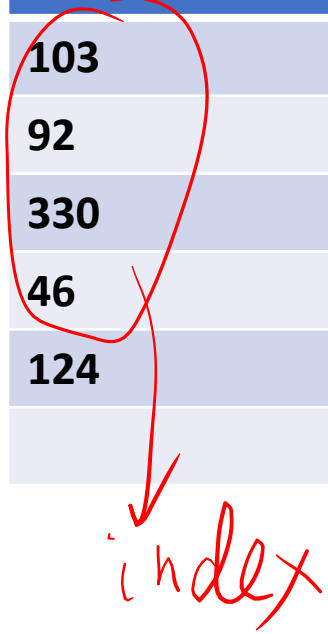
Hashing

Locker Number	Name
103	Rick
92	Kiri
330	Mary Catherine
46	Blake
124	Erin

Hashing

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index



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index

[[1, 2, 3], ..., ["kivi"]]

Hashing

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index Locker[92]

[[[[] - - - ["kiri"]]]

01 23 92

Hashing

O(1) ✓

Locker Number	Name
103	Rick
92	Kiri
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index

Locker[92]

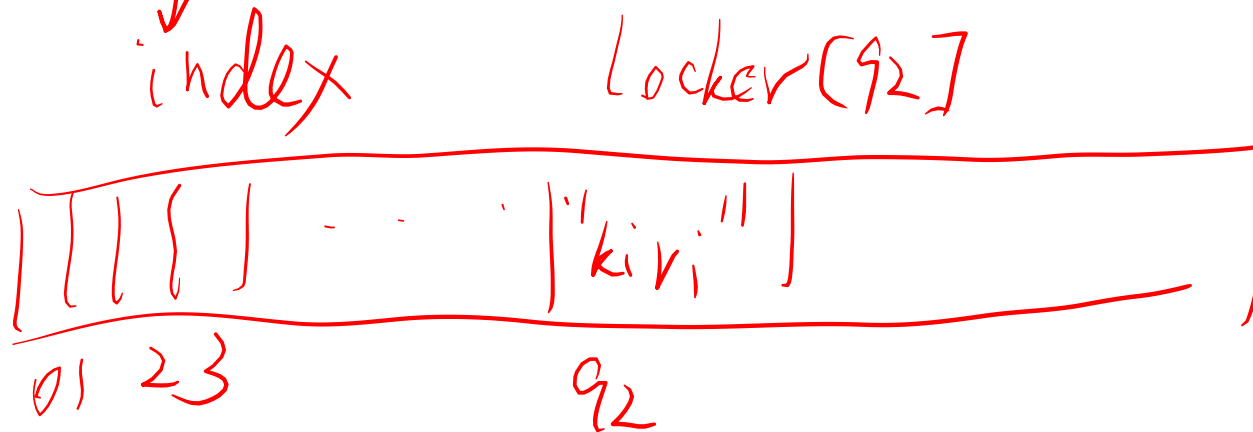
[[] [] [] ... ["kiri"]]
0 1 2 3 92

Hashing

Locker Number	Name
103	Rick
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$O(1)$ ✓

$O(X)$
keyspace



Hashing

Commonly:

Just use a “hash table” to implement a _____.

Hashing

Commonly:

Just use a “hash table” to implement a dictionary.

Dictionary ADT

Data is often organized into key/value pairs:

UIN → Advising Record

Course Number → Lecture/Lab Schedule

Node → Incident Edges

Flight Number → Arrival Information

URL → HTML Page

...

Dictionary.h

```
1  #ifndef DICTIONARY_H
2  #define DICTIONARY_H
3
4  template <class K, class V>
5  class Dictionary {
6      public:
7          void insert(K & k, V & v);
8          void remove(const K & k);
9          V & find(const K & k) const;
10
11
12
13
14
15
16      private:
17  };
18
19  #endif
20
21
22
```

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associative array []

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associative array []
V & operator[] (const K & k) const;

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```

associative array []
V & operator[] (const K & k) const;
d["wade"]

Hashing

Goals:

We want to define a **keyspace**, a (mathematical) description of the keys for a set of data.

...use a function to map the **keyspace** into a small set of integers.

Hashing

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...use a function to map the **keyspace** into a small set of integers.

$$h: K \rightarrow N$$